Opening the Black Box*

A Theory of the Value of Data in Prediction

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Abstract

This paper develops a theory of the value of data for prediction. A platform predicts a target variable for a target individual. To improve its prediction, the platform can collect covariates on that individual and training observations from similar individuals, where each observation contains the target variable and a possibly different set of covariates. The model delivers two main results. First, covariates exhibit economies of scope: one covariate becomes more valuable when the platform observes others. Second, covariates and observations are complements in small datasets but can become substitutes in large datasets. I then apply the model to competition for covariates between two ecosystems, showing how economies of scope can create a data-feedback force in app competition.

JEL Classification: C11, D83, L12, L40.

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1 Introduction

In digital markets, data matter because they help firms predict user behavior. ChatGPT and Gemini predict which response a user will find helpful given a prompt and context. Amazon and Uber predict where demand will arise. Google and Meta predict which ad a user will click. Netflix predicts which content a user will enjoy.

Over the last decade, a few firms have come to dominate many digital markets, and these firms often control vast amounts of data. Competition authorities have started to investigate whether data can entrench incumbents. The 2022 U.S. House Report warns that "data advantages [...] can reinforce dominance and serve as a barrier to entry."[1] Similar concerns appear in the EU Data Act,[2] and in major policy reports.[3]

Industry leaders once made similar claims. As Google's CEO stated in 2009, "Scale is the key. We just have so much scale in terms of the data we can bring to bear."[4] Subsequently, however, technology firms have pushed back, arguing that data exhibit diminishing returns to scale by the Law of Large Numbers, and that data concentration in the hands of a few large firms reflects superior technology rather than entry barriers.[5]

To evaluate these competing views, we must distinguish two ways firms acquire data. A firm can observe more individuals, increasing sample size, or it can observe more attributes of each individual, increasing covariate scope. This paper asks how each margin improves prediction. Empirically and conceptually, these two channels—*economies of scale* and *economies of scope*—are often conflated. I disentangle them, showing that once economies of scope are considered, data may have increasing returns and thereby naturally concentrate in the hands of one firm.

Theory: the value of data. I develop a Bayesian model that separates these two dimensions. A firm predicts a target variable for a target individual—for example, whether a user will click on a targeted ad. The target variable depends on many potential covariates, such as past searches, browsing behavior, and demographic characteristics. Each covariate affects the outcome through an unknown weight.

The firm starts with prior uncertainty about these weights. It can collect two types of data. First, it can collect *training data*: covariates and target variables for a sample of individuals. These data help the firm learn how covariates relate to outcomes. Second, it can collect *target covariates*: covariates of the target individual. These data let the firm apply what it has learned

[1] House Committee Print 117-8 "Investigation of Competition in Digital Markets, Majority Staff Report and Recommendations", 2022.

[2] Regulation (EU) 2023/2854 of the European Parliament and of the Council of 13 December 2023 on harmonized rules on fair access to and use of data and amending Regulation (EU) 2017/2394 and Directive (EU) 2020/1828 (Data Act). The regulation applies from 12 September 2025.

[3] Digital Platforms (2019) and UK Competition and Markets Authority (2019).

[4] Schmidt (2009), "How Google Plans to Stay Ahead in Search," Bloomberg, 2 October.

[5] (Varian 2015; Varian 2018; Bajari et al. 2019).

to the individual it wants to predict.

Thus the firm designs a dataset by choosing three objects: (i) the number of training observations N; (ii) the set of *training covariates* $\mathcal{T}$ observed in the training data; and (iii) the set of *target covariates* $\mathcal{T}_0$ observed for the target individual. Because each object is costly, the firm trades off better prediction against higher data-collection costs.

For any dataset design, I derive the optimal Bayesian predictor and define the value of data as the expected reduction in out-of-sample mean squared error. I show that this value is multiplicatively separable: it is the product of how much the training data teach the firm about the weights and how many target covariates the firm can apply them to. This separation is what lets me treat scale, training scope, and target scope as three distinct economic margins rather than a single undifferentiated amount of data.

I then derive a closed-form expression for dataset value in the high-dimensional limit, for sequences of asymptotically non-degenerate designs. Dataset value rises for three reasons. More training covariates reduce residual noise. More observations estimate the weights more precisely. More target covariates let the firm apply more of these estimates to the target individual.

Two results follow. First, training covariates exhibit *economies of scope*. The marginal value of an additional training covariate is higher when the firm already observes more training covariates, because broader training scope reduces residual noise and improves learning about all trained weights. This is precisely the margin the diminishing returns argument based on the Law of Large Numbers overlooks: it holds the covariate set fixed and adds observations, whereas economies of scope operate when the firm widens the covariate set itself.

Second, training covariates and observations are *complements* in small datasets but can become *substitutes* in large datasets. In a small dataset, an additional covariate raises the marginal value of an observation. In a large dataset, additional covariates mainly reduce residual noise and can reduce the marginal willingness to pay for further noise reductions, lowering the marginal value of further observations. The firms most likely to treat observations and covariates as substitutes are therefore the largest — those already observing many users. This substitution result disappears when idiosyncratic noise is sufficiently large and there is high value to reducing noise. Therefore, substitutability can emerge only in relatively "easy" prediction problems, in which the prediction problem has low noise.[6]

These curvature results clarify when data combination creates value. Since value is concave in scale but convex in scope, the effect of a merger depends on the margin it expands. Pooling the same covariates across more users adds scale and is therefore strictly subadditive: the merged dataset is worth less than the two datasets separately. Pooling distinct covariates for the same users adds scope and is strictly superadditive: the merged dataset creates genuine synergies. The same scale–scope logic also predicts which firms will bid most aggressively

[6] I also use simulations to test whether these results survive non-Gaussian shocks and departures from independent covariates, see Appendix B.

for covariate-rich data. The model therefore gives merger review sharp, testable predictions from the statistical structure of prediction alone, without relying on strategic interaction.

Application: ecosystem competition for apps. I use the model to study competition between two ecosystems that make predictions independently. Each ecosystem has a fixed mass of captive users, who generate observations, and competes for apps, which generate covariates. Ecosystems first choose *ad valorem* fees to attract apps. After app demand realizes, each ecosystem chooses how many acquired covariates to deploy as target covariates and pays a deployment cost.

Economies of scope across covariates create a data-feedback force in app competition. When prediction value is low, the data-generating process is noisy, the captive user base is small, or apps are sufficiently differentiated, this force is weak. The market then admits a unique, locally stable symmetric equilibrium. When prediction value is high, app data are rich, and apps are weakly differentiated, the same force can make the symmetric equilibrium locally unstable. A larger captive user base can strengthen this instability because additional users make app-generated covariates more valuable. The effect is strongest when observations and app scope are complementary; it fades once additional observations have mostly exhausted their learning value.

The results show that asymmetries and rich-get-richer effects emerge endogenously from the nonconvex way data generate value in the model.

Contribution and Related literature. The paper contributes to four strands of literature. First, the paper contributes to the literature on economies of scale and scope in data. Most models fix the set of covariates and study returns to observations, as in Varian (2015), Varian (2018), and Bajari et al. (2019). I instead let firms choose which covariates to collect. The framework therefore rationalizes empirical evidence on complementarities and economies of scope in Schaefer and Sapi (2023) and Carballa-Smichowski et al. (2025a).

Schaefer (2025) develops a complementary frequentist approach and shows that the eigenspectrum of the covariate matrix shapes returns to scale, consistent with my comparative statics. Lee and Wright (2023) find diminishing returns to sample size but complementarities between sample size and covariates in a recommender system. Several theory papers derive increasing returns to data from learning dynamics (Hagiu and Wright 2023; Prüfer and Schottmüller 2021; Farboodi and Veldkamp 2025; Aral et al. 2008). I complement this work by identifying a static source of increasing returns.

These curvature properties also map directly into the value of data mergers, connecting my results to Gu et al. (2021) and Cornière and Taylor (2024). The model distinguishes mergers that pool observations from mergers that pool covariates. Observation-pooling mergers do not generate an efficiency defense, because scale exhibits diminishing returns. Covariate-pooling mergers can generate one, because they exploit economies of scope. The paper there-

fore connects the statistical structure of prediction to the efficiency defense in merger review.

Secondly, it contributes to the information-design literature on the value of data, starting with Bergemann et al. (2018) and continuing with Jones and Tonetti (2020), Bergemann et al. (2022), Bergemann and Bonatti (2024), and Acemoğlu et al. (2022). These papers study how much an agent is willing to pay for more informative experiments by modeling the choice of an information structure. My setting is more structured than a Blackwell experiment. That structure lets me distinguish observations, training covariates, and target covariates. It also lets me map the value of data to transparent statistical objects: covariate variances, posterior variances, and the ridge regression estimator.

Thirdly, the model's application contributes to the literature on digital ecosystems. Several papers build on the assumption that there are economies of scope (also called data-driven network effects) between data generated by apps providing different services Carballa-Smichowski et al. (2025b), De Corniere and Taylor (2025), Calzolari et al. (2025), and Rhodes et al. (2026). I contribute to this literature by microfounding these economies of scope and shedding light on the statistical properties of the data which make them strongest. Moreover, in the application I show how statistical economies of scope in covariates alone can cause tipping and data concentration, leading to asymmetries in ecosystems that compete to attract apps.

Lastly, the paper is related to the literature on competing prediction models pioneered by Montiel Olea et al. (2022), Iyer and Ke (2024), and Dasaratha et al. (2025), who study competition between models with different covariates. By contrast, I model covariates and observations jointly and separate training data from prediction data. This separation generates structural non-convexities and complementarities in the value of data. Finally, Strzalecki (2024) interprets deviations from Bayesian updating through a primitive misspecification parameter. In my framework, a related distortion arises endogenously from standard Bayesian behavior under incomplete data collection.

Roadmap. Section 2 introduces the data-generating process and the firm's dataset-design problem. Section 3 derives the optimal predictor, characterizes the value of data, and studies returns to scale, scope, and complementarities across design margins. Section 4 analyzes ecosystem competition for apps. Section 5 concludes. Appendix A contains the proofs. Appendix B contains the simulations.

2 Model

2.1 Setup

An agent (she) seeks to predict a scalar target variable y_0 for a target individual indexed by 0 and drawn from a countably infinite population $\mathbb{N}_0$. To do so, she can collect outcomes

and a chosen set of covariates for a training sample, together with a possibly different set of covariates for the target individual. The agent chooses the sample size and which covariates to observe, trading off increases in prediction accuracy and data-collection costs.

2.1.1 Data-Generating Process

The environment consists of a data-generating process (DGP) where the set of potential covariates is $\mathcal{K} \equiv \{1, ..., K\}$. For each individual $i \in \mathbb{N}_0$, the target variable is a linear combination

$$y_i \;=\; \boldsymbol{x}_i' \boldsymbol{\beta} + \eta_i,$$

where $\boldsymbol{x}_i \in \mathbb{R}^K$ is the covariate vector of individual i, $\boldsymbol{\beta} \in \mathbb{R}^K$ is a vector of unknown weights, and $\eta_i \in \mathbb{R}$ is an idiosyncratic error.

The weights are independent of the covariates and the error and are mutually independent zero-mean gaussians: $\boldsymbol{\beta} \sim \mathcal{N}\left(0, \sigma_\beta^2 \cdot \boldsymbol{I}_K\right)$, where $\sigma_\beta^2 > 0$ reflects prior uncertainty about the weights. Covariates are i.i.d. across individuals and are mutually independent zero-mean gaussians: $\boldsymbol{x}_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \sigma_x^2/K \cdot \boldsymbol{I}_K\right)$, where $\sigma_x^2 > 0$ reflects the signal contained in the covariates.[7] The idiosyncratic errors are i.i.d. across individuals and Gaussian with mean zero: $\eta_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2 \sigma_\beta^2)$, where $\sigma_\eta^2 > 0$ reflects the (scaled) idiosyncratic noise.

Under these assumptions, $\text{Var}[y_i] = \sigma_\beta^2(\sigma_x^2 + \sigma_\eta^2)$, which is equal for all individuals and does not depend on K. The prior is the maximal-entropy distribution consistent with the second-moments.[8]

2.1.2 Dataset Design and Predictor

The agent chooses a ***dataset design***, which consists of a triple $\mathcal{D} \equiv (N, \mathcal{T}, \mathcal{T}_0)$, collecting three objects: (i) a sample size $N \in \mathbb{N}_0$;[9] (ii) a set of training covariates, $\mathcal{T} \subseteq \mathcal{K}$, observed on the N individuals in the sample, with $T \equiv |\mathcal{T}|$; and (iii) a set of target covariates, $\mathcal{T}_0 \subseteq \mathcal{K}$, observed on the target individual, with $T_0 \equiv |\mathcal{T}_0|$. A design induces a ***dataset***, which is a realization $\boldsymbol{d}_D \equiv (\{(y_i, \boldsymbol{x}_{i,\mathcal{T}})\}_{i=1}^N, \boldsymbol{x}_{0,\mathcal{T}_0}) \in \mathbb{R}^{N\times(1+T)} \times \mathbb{R}^{T_0}$, where for any vector $\boldsymbol{v} \in \mathbb{R}^K$ and index set $\mathcal{K}' \subseteq \mathcal{K}$, I let $\boldsymbol{v}_{\mathcal{K}'} \equiv (v_k)_{k\in\mathcal{K}'}$ denote the corresponding subvector. I denote the *empty design* by $\varnothing$.

Given a design $\mathcal{D}$, the agent chooses a ***predictor***, which is a measurable map $\hat{y}_D : \mathbb{R}^{N\times(1+T)} \times \mathbb{R}^{T_0} \to \mathbb{R}$ which assigns a prediction to each realized dataset.

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[7] The scaling by $1/K$ keeps the total covariate variance $\sum_{k=1}^K \text{Var}(x_{ik})$ equal to σ_x^2, independently of K.

[8] As the variance of covariates/weights is identical, the agent ignores which covariates have a greater influence on the target variable. Similarly, as the weights have zero mean, she does not know whether a covariate affects the target positively or negatively. Furthermore, as the covariates are independent, she does not know whether any two covariates are positively or negatively correlated. Finally, as all the DGP's components are gaussian, she has the maximum entropy distribution, for given first and second moments.

[9] Because the variables are i.i.d. across individuals, there is no loss of generality in assuming the sample is $\{1, \dots, N\} \subset \mathbb{N}_0$.

2.1.3 Loss, Design Value and Cost

The agent faces a quadratic loss function $\ell(y_0, \hat{y}) = (y_0 - \hat{y})^2$, for a given prediction $\hat{y} \in \mathbb{R}$. I assume the choice of predictor is costless, so the agent uses the ***optimal predictor***

$$\hat{y}_D^*(\boldsymbol{d}) \equiv \arg\min_{\hat{y}_D} \underbrace{\mathbb{E}\left[\ell(y_0, \hat{y}_D(\boldsymbol{d})) \mid \boldsymbol{d}\right]}_{\text{posterior risk}}.^{10}$$

The dataset $\boldsymbol{d}_D$ is distributed according to the truncation of the DGP induced by $\boldsymbol{D}$, which I denote by P_D.

Given a realized dataset $\boldsymbol{d}$ induced by design $\boldsymbol{D}$, define its *realized value* as

$$\hat{v}_D(\boldsymbol{d}) \equiv \mathrm{Var}(y_0) - \mathrm{Var}(y_0 \mid \boldsymbol{d}).$$

This is the reduction in posterior risk achieved after observing the particular dataset $\boldsymbol{d}$.

The *value of a design* is the ex ante expectation of this realized value:

$$v(\boldsymbol{D}) \equiv \mathbb{E}_{\boldsymbol{d}_D \sim P_D}\left[\hat{v}_D(\boldsymbol{d}_D)\right] = \mathrm{Var}(y_0) - \mathbb{E}_{\boldsymbol{d}_D \sim P_D}\left[\mathrm{Var}(y_0 \mid \boldsymbol{d}_D)\right].$$

A design's cost depends only on its relative dimensions:

$$C(\boldsymbol{D}) = C(N/K, T/K, T_0/K),$$

where C is weakly increasing in each argument and strictly increasing in at least one argument, with $C(0, 0, 0) = 0$. Thus, to maximize its payoff, the agent solves

$$\max_{\boldsymbol{D}} \; v(\boldsymbol{D}) - C(\boldsymbol{D}). \tag{1}$$

3 Analysis

3.1 Optimal Predictor

Under quadratic loss, the optimal predictor is the posterior mean of y_0 given the dataset.

Proposition 1 (Optimal predictor)**.** *The optimal predictor is*

$$\hat{y}_D^*(\boldsymbol{d}) = \mathbb{E}\left[y_0 \mid \boldsymbol{d}\right] = \boldsymbol{x}_{0,\mathcal{T}_0 \cap \mathcal{T}}^\top \mathbb{E}[\boldsymbol{\beta}_{\mathcal{T}_0 \cap \mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})].$$

Two implications are immediate. First, target variables are conditionally independent across individuals given $\boldsymbol{\beta}$, so the predictor depends on the training sample only through the posterior of the weights. Second, only covariates in the intersection $\mathcal{T}_0 \cap \mathcal{T}$ enter the predictor: a target covariate not in $\mathcal{T}$ has posterior mean zero by symmetry, while a training

[10] Quadratic loss ensures the optimal predictor exists and is unique.

covariate not in $\mathcal{T}_0$ is unobserved at prediction time. The value of a design therefore reduces to the expected variance reduction it generates:

$$v(\mathcal{D}) = \mathrm{Var}\left[y_0\right] - \mathbb{E}_{\boldsymbol{d}_D \sim P_D} \mathrm{Var}\left[y_0 \mid \boldsymbol{d}_D\right].$$

3.2 Nested Designs and the High-Dimensional Limit

Because target covariates excluded from training contribute nothing to the predictor and are weakly costly, any design $(N, \mathcal{T}, \mathcal{T}_0)$ is weakly dominated by $(N, \mathcal{T}, \mathcal{T}_0 \cap \mathcal{T})$. I therefore restrict attention to nested designs with $\mathcal{T}_0 \subseteq \mathcal{T}$. Under index-wise symmetry, only the number of covariates matters, so any such design can be represented as $\mathcal{D} = (N, T, T_0)$ with $T_0 \leq T$.

Given K and conditional on the resulting dataset design $\mathcal{D}(K)$, for any individual i in the training sample, we can decompose the target variable as

$$y_i^{(K)} = \sum_{k=1}^{T(K)} \beta_k^{(K)} x_{ik}^{(K)} + \varepsilon_i^{(K)}, \text{ with } \quad \varepsilon_i^{(K)} \equiv \sum_{k=T(K)+1}^{K} \beta_k^{(K)} x_{ik}^{(K)} + \eta_i^{(K)},$$

where $\varepsilon_i^{(K)}$ is the *training residual* due to the omitted covariates and the idiosyncratic error $\eta_i^{(K)}$.

To obtain tractable expressions, I work in the high-dimensional limit considering a sequence of designs indexed by K.

Assumption 1 (Sequence of designs). *The sequence* $\{D(K)\}_{K \in \mathbb{N}_+}$ *admits constants* $(n, t, t_0) \in (0, \infty) \times (0, 1]^2$ *such that, as* $K \to \infty$,

$$\frac{N(K)}{K} \to n, \qquad \frac{T(K)}{K} \to t, \qquad \frac{T_0(K)}{K} \to t_0.$$

I refer to (n, t, t_0) as the *design margins*: n is the *(training) scale*, t the *training scope*, and t_0 the *target scope*. The assumption is standard in random matrix theory and reflects the data-rich environments characteristic of modern machine learning.[11]

Conditional on the omitted coefficients, the residual vector is Gaussian:

$$\boldsymbol{\varepsilon}_{N(K)}^{(K)} \mid \boldsymbol{\beta}_{\mathcal{T}(K)^c} \sim \mathcal{N}\left(0,\ \sigma_\beta^2 \Sigma_K^2 \boldsymbol{I}_{N(K)}\right),$$

where

$$\Sigma_K^2 \equiv \sigma_\eta^2 + \frac{\sigma_x^2}{K \sigma_\beta^2} \sum_{k=T(K)+1}^{K} \left(\beta_k^{(K)}\right)^2.$$

By the law of large numbers,

$$\Sigma_K^2 \xrightarrow{p} \sigma_\varepsilon^2(t) \equiv (1-t)\sigma_x^2 + \sigma_\eta^2.$$

[11] See Marčenko and Pastur (1967), Bai and Silverstein (2009), and Hastie et al. (2020). The novelty here is the joint control of three margins.

Thus the residual variance generated by omitted covariates concentrates around a deterministic limit. The closed form below uses the corresponding posterior trace equivalence: the average posterior variance of the trained weights has the same limit as in a Gaussian linear model with residual variance $\sigma_\beta^2\sigma_\varepsilon^2(t)$. A formal statement is given in Appendix A.

3.3 The Value of a Dataset Design

Define the *(relative) information on trained weights* as the asymptotic share of prior weight variance learned in training:

$$\mathcal{I}(n,t) \equiv \lim_{K\to\infty} \frac{\operatorname{tr}\left[\operatorname{Var}[\boldsymbol{\beta}_{\mathcal{T}(K)}] - \operatorname{Var}\left[\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right]\right]}{\operatorname{tr}[\operatorname{Var}[\boldsymbol{\beta}_{\mathcal{T}(K)}]]}. \tag{2}$$

Thus $\mathcal{I}(n,t) = 0$ means training is uninformative, while $\mathcal{I}(n,t) = 1$ means weights are learned exactly.

Theorem 1 (Asymptotic value of a design)**.** *Under Assumption 1, for any nested sequence of designs* $\{D(K)\}_{K\in\mathbb{N}_+}$*,*

$$\hat{v}_{D(K)}(\boldsymbol{d}_{D(K)}) \xrightarrow{p} \mathcal{V}(n,t,t_0),$$

where

$$\mathcal{V}(n,t,t_0) = \sigma_\beta^2\sigma_x^2 t_0 \mathcal{I}(n,t).$$

Consequently,

$$v(D(K)) \longrightarrow \mathcal{V}(n,t,t_0).$$

Proof. The proof hinges on Lemma 4 and is presented in Appendix A. □

The closed form arises because the empirical spectral distribution of $\boldsymbol{X}'_{N,T}\boldsymbol{X}_{N,T}$ converges to a Marchenko–Pastur law, which admits explicit eigenvalue moments under independence and homoskedasticity.[12]

Decomposition. The value of a dataset is the product of two terms with distinct economic interpretations. The *target signal* $\sigma_x^2 t_0$ is the part of target-variable variance attributable to covariates that are both observed at prediction time and used in training. The *weight information* $\sigma_\beta^2\mathcal{I}(n,t)$ is the resolution of prior uncertainty about the predictor weights achieved by training data of scale n and scope t.

If any margin is zero, the dataset has no value: $\mathcal{V}(0,t,t_0) = \mathcal{V}(n,0,t_0) = \mathcal{V}(n,t,0) = 0$. Without observations or training scope, the agent cannot learn weights; without target scope, learned weights cannot be applied. Around the origin, the three margins are therefore

[12]Section B verifies via simulation that the formula is robust to local correlation and heavy tails, and characterizes when heteroskedasticity matters.

hard complements. As scale grows, weights are learned almost perfectly: $\lim_{n\to\infty} \mathcal{V}(n, t, t_0) = \sigma_\beta^2 \sigma_x^2 t_0$, and the same hard-complementarity pattern between t and t_0 reemerges at the limit.[13]

3.4 Signal-Share Representation

The closed form admits a fixed-point representation that grounds the value of data in observable sample statistics. For each K, define the empirical training signal

$$\mathcal{S}_K \equiv \frac{1}{T(K)} \operatorname{tr}\left(\boldsymbol{X}_{N(K),\mathcal{T}(K)}^\top \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right),$$

and, for the target individual,

$$u_0^{(K)} \equiv y_0^{(K)} - \boldsymbol{x}_{0,\mathcal{T}(K)}^\top \mathbb{E}\left[\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right], \qquad \mathcal{E}_K \equiv \frac{1}{\sigma_\beta^2} \operatorname{Var}\left(u_0^{(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right).$$

The scalar $\mathcal{S}_K$ measures the average signal contained in the training matrix, while $\mathcal{E}_K$ is the normalized conditional variance of the realized prediction residual.

Proposition 2 (Signal-share representation)**.** *As $K \to \infty$,*

$$\mathcal{S}_K \xrightarrow{p} \mathcal{S}(n), \qquad \mathcal{E}_K \xrightarrow{p} \mathcal{E}(\mathcal{I}(n,t); t),$$

where

$$\mathcal{S}(n) \equiv n\sigma_x^2, \qquad \mathcal{E}(\mathcal{I}; t) \equiv \sigma_\varepsilon^2(t) + t\sigma_x^2(1 - \mathcal{I}).$$

Moreover, $\mathcal{I}(n, t)$ is the unique element of $[0, 1]$ satisfying

$$\mathcal{I}(n,t) = \Psi(\mathcal{I}(n,t); n, t), \qquad \Psi(\mathcal{I}; n, t) \equiv \frac{\mathcal{S}(n)}{\mathcal{S}(n) + \mathcal{E}(\mathcal{I}; t)}.$$

Weight information is therefore the unique self-consistent signal share in the training problem. The training signal $\mathcal{S}(n)$ is linear in the scale n. The effective training noise $\mathcal{E}(\mathcal{I}; t)$ has two components: misspecification noise $\sigma_\varepsilon^2(t)$, decreasing in t, and estimation noise $t\sigma_x^2(1 - \mathcal{I})$, which captures residual posterior uncertainty about the trained weights and is increasing in the number of weights estimated. The first effect dominates, so $\mathcal{E}(\mathcal{I}; t)$ is decreasing in t and both training scope and training scale raise weight information.

[13] When $t = 0$, feasibility implies $t_0 = 0$, so total value is zero. The formula should therefore be read on the feasible set $0 \le t_0 \le t$: while $\mathcal{I}(n, t)$ has a positive limit as $t \downarrow 0$, it is a per-trained-weight information share, and no weights can be deployed when $t_0 = 0$.

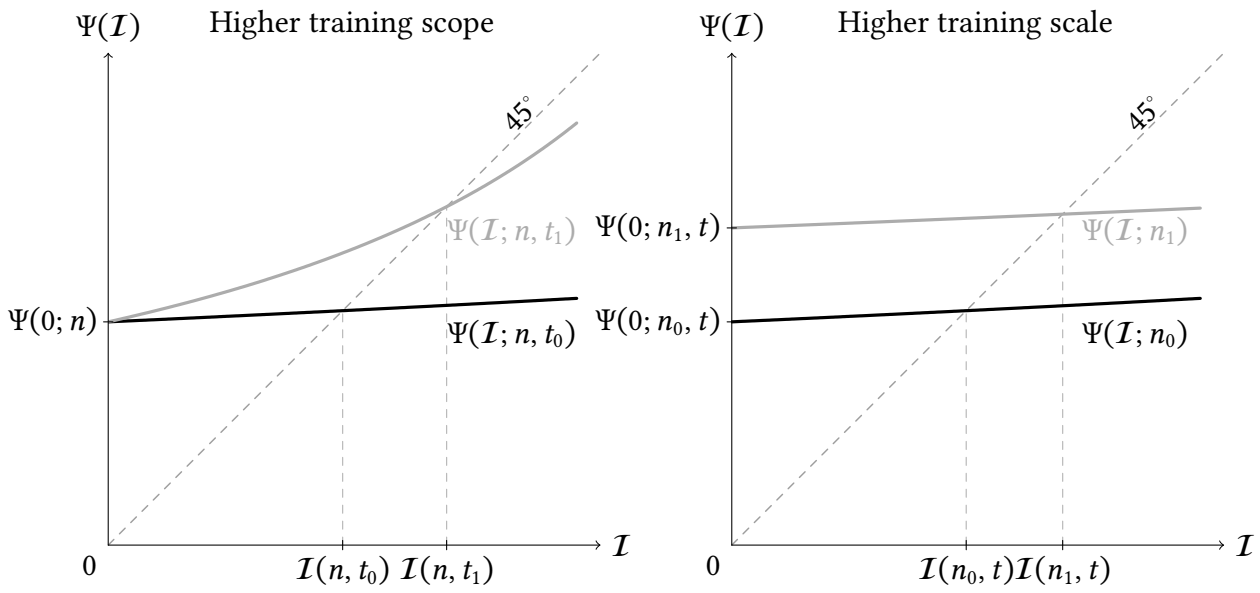


Figure 1: Weight information as training signal share. In each panel, $\mathcal{I}(n,t)$ is the intersection between $\Psi(\mathcal{I})$ and the 45° line. Left: holding training scale n fixed, a higher training scope t steepens $\Psi(\mathcal{I})$ without changing $\Psi(0)$. Right: holding training scope t fixed, a higher training scale n shifts $\Psi(\mathcal{I})$ upward.

The two design margins act on the fixed point through qualitatively different channels, as Figure 1 illustrates. Training scale shifts the curve $\Psi(\mathcal{I};n,t)$ upward by raising the signal $S(n)$, producing a level effect. Training scope steepens the curve by amplifying the dependence of noise on $\mathcal{I}$ itself: because the noise term contains $-t\sigma_x^2\mathcal{I}$, broader scope makes the same level of weight information generate a larger reduction in residual noise. Scope thus operates through a feedback channel that scale does not. This asymmetry — scope as amplification, scale as level — is the source of the regime-dependent comparative statics in Proposition 4 below.

3.5 Returns to Design Margins

Let $\xi \equiv \sigma_\eta^2/\sigma_x^2$ denote the noise-to-signal ratio. Normalizing $\sigma_x^2 = 1$ and dividing by σ_β^2, the relative value function is

$$V(n,t,t_0) \equiv \frac{\mathcal{V}(n,t,t_0)}{\sigma_\beta^2} = t_0 \cdot \mathcal{I}(n,t).$$

V is linear in t_0 with slope $\mathcal{I}(n,t)$: increasing t_0 enlarges the set of weights that are learned and used in prediction.

3.5.1 Diminishing returns to scale.

Corollary 1 (Returns to scale). *V is strictly increasing and strictly concave in n:*

$$\frac{\partial V}{\partial n} = t_0 \frac{\partial \mathcal{I}}{\partial n} > 0, \qquad \frac{\partial^2 V}{\partial n^2} = t_0 \frac{\partial^2 \mathcal{I}}{\partial n^2} < 0.$$

The result follows from the linearity of the training signal in n and the concavity of $\mathcal{I}$ in $S(n)$. Each additional observation refines the posterior, but does so by less than the previous one. The pattern is consistent with empirical evidence in Bajari et al. (2019), Goldfarb and Tucker (2011), Schaefer and Sapi (2023), and Lee and Wright (2023).

3.5.2 Economies of scope in training.

Proposition 3 (Economies of training scope). *V is strictly increasing and strictly convex in t:*

$$\frac{\partial V}{\partial t} = t_0 \frac{\partial \mathcal{I}}{\partial t} > 0, \qquad \frac{\partial^2 V}{\partial t^2} = t_0 \frac{\partial^2 \mathcal{I}}{\partial t^2} > 0.$$

Training scope raises value because broader scope improves the precision of the deployed weights by reducing residual noise. The convexity comes from the same fixed-point structure that produced Proposition 2. Broader scope reduces the misspecification component of noise, and the information so generated feeds back into further reductions because the noise term itself depends on $\mathcal{I}$. Each additional covariate is therefore more valuable than the last. This economies-of-scope property is consistent with the empirical findings in Carballa-Smichowski et al. (2025a).

3.5.3 Are observations and covariates complements or substitutes?

Proposition 4 (Complements and substitutes). *Let $V(n, t, t_0) = t_0 \mathcal{I}(n, t)$, with $n > 0$ and $0 < t_0 \leq t \leq 1$. Then*

$$V_{nt_0} > 0, \qquad V_{tt_0} > 0.$$

For each $t \in (0, 1]$, there exists a threshold $\hat{n}(t; \xi) > 0$ such that

$$V_{nt} \gtreqless 0 \quad \Longleftrightarrow \quad n \lesseqgtr \hat{n}(t; \xi).$$

The threshold is decreasing in t and increasing in ξ.

Since $V_{tt_0} > 0$, target scope complements both training margins. By contrast, training scale and training scope are complements only below the threshold $\hat{n}(t; \xi)$. Above that threshold, they are substitutes. Higher idiosyncratic noise raises $\hat{n}(t; \xi)$ and therefore expands the complementarity region.

The interaction between training scale and training scope is subtler, because scope reduces noise through two opposing channels. The first channel is direct: a larger t lowers $\sigma_\varepsilon^2(t)$,

which reduces the marginal value of an additional observation because the learning problem is already less noisy. This force pushes toward substitutability. The second channel is indirect: a larger t amplifies the feedback from existing weight information into further noise reduction, through the $-t\mathcal{I}$ term in $\mathcal{E}$. This force pushes toward complementarity.

The result sheds light on the complementarities between observations and covariates identified empirically in Schaefer and Sapi (2023) and Lee and Wright (2023), showing that they stem from the fact that more observations allow the learning to pin down better the parameters (which are equal in number to the covariates). Moreover, it shows that this complementarity is not a general property and observations and covariates can behave as substitutes in large datasets.

These two statistical properties—economies of scope in Proposition 3 and the regime-dependent interaction between scale and scope in Proposition 4—drive the economic mechanisms developed in the rest of the paper.

3.5.4 Data combination: mergers and acquisitions

The comparative statics above describe how a single agent's value changes with its own design margins. The same curvature also determines the value of combining datasets held by different agents. Since value is concave in scale and convex in scope, a merged dataset need not be worth the sum of the standalone datasets. The sign of the gap depends on which margin the merger expands.

This gives a simple diagnostic for merger review. One need not model strategic interaction to ask whether a data combination creates value. It is enough to ask whether the merger pools observations on the same covariates or pools different covariates on the same users. Proposition 5 isolates these two polar cases.

Proposition 5 (Data mergers: composition determines value)**.** *Fix ξ and consider a merger between two data holders.*

(i) Observation pooling. *Suppose the two data holders observe the same covariate set of scope t on disjoint samples of sizes $n_1, n_2 > 0$. If target scope $t_0 > 0$ is held fixed, the merger is strictly subadditive:*

$$t_0\, \mathcal{I}(n_1 + n_2, t) \;<\; t_0\, \mathcal{I}(n_1, t) + t_0\, \mathcal{I}(n_2, t).$$

(ii) Covariate pooling. *Suppose the two data holders observe the same sample of size n on disjoint covariate sets of scopes $t_1, t_2 > 0$, with $t_1 + t_2 \leq 1$. If each data holder deploys its full scope, the merger is strictly superadditive:*

$$(t_1 + t_2)\, \mathcal{I}(n, t_1 + t_2) \;>\; t_1\, \mathcal{I}(n, t_1) + t_2\, \mathcal{I}(n, t_2).$$

Proposition 5 translates the curvature results into merger language. Part (i) is diminishing returns to scale (Corollary 1). When firms pool more observations of the same covariates, the concavity of $\mathcal{I}$ in n makes the merged dataset worth strictly less than the two standalone datasets. Part (ii) is economies of scope (Proposition 3). When firms pool different covariates on the same users, the convexity of $s \mapsto s\,\mathcal{I}(n, s)$ makes the merged dataset worth strictly more than the standalone sum.

Thus the dimension of overlap determines the direction of the efficiency defense. Mergers that consolidate the same attributes across more users face diminishing returns. Mergers that combine distinct attributes on the same users are the ones that generate genuine data synergies.

Proposition 5 asks whether combining data creates value. The next result asks who values the acquisition most. The relevant object is no longer curvature in a single margin, but the interaction between scale and scope characterized in Proposition 4.

Corollary 2 (Acquisition incentives for covariate-rich targets)**.** *Consider a buyer with scale n and training scope t. Suppose it can absorb an additional mass Δt of covariates, observed on its own users, from a covariate-rich target. To first order, the buyer's value gain is proportional to*

$$\mathcal{I}_t(n, t)\,\Delta t.$$

How this gain varies with the buyer's own scale is determined by

$$\frac{\partial}{\partial n}\mathcal{I}_t(n, t) = \mathcal{I}_{nt}(n, t) \;\gtreqless\; 0 \quad \Longleftrightarrow \quad n \;\lesseqgtr\; \widehat{n}(t; \xi).$$

Therefore, when $n < \widehat{n}(t; \xi)$, larger buyers value the covariate acquisition more than smaller rivals do. When $n > \widehat{n}(t; \xi)$, larger buyers value it less. The range in which scale-rich buyers bid more aggressively expands with the noise-to-signal ratio ξ and shrinks with the buyer's existing scope t.

Together, Proposition 5 and Corollary 2 show that the statistical primitives of Section 3.5 already imply sharp predictions for merger and acquisition policy, even without strategic interaction. Diminishing returns to scale limit the value of pooling more users with the same attributes. Economies of scope raise the value of combining different attributes on the same users. The interaction between scale and scope then determines whether large incumbents or smaller rivals have the stronger acquisition motive.

The remainder of the paper embeds these same forces in a model of ecosystem competition. There, economies of scope operate not through a single firm's merger calculus, but through the equilibrium allocation of apps across competing platforms.

4 Ecosystem Competition for Covariates

Two ecosystems, indexed by $i \in \{A, B\}$, each have n captive users, who supply observations (training scale), and compete for *apps*, who supply covariates (training scope). Each ecosystem uses the data it acquires internally to produce a prediction of value to its own operations (e.g. ad targeting, recommendation, demand forecasting).

Apps. A unit mass of apps is uniformly distributed on a Hotelling segment $[0, 1]$, with A and B again at the endpoints. An app of type $\vartheta \in [0, 1]$ joining ecosystem i pays an ad-valorem fee $f_i \in \mathbb{R}$ (which can be negative) and bears a symmetric transportation cost. Throughout, I work in per-app revenue $p_i \equiv \varrho f_i \in \mathbb{R}$ rather than the fee f_i; the two are equivalent and p yields cleaner expressions. Thus an app of type ϑ chooses among ecosystem A, ecosystem B, and non-participation, with utilities

$$u_A(\vartheta) = \varrho - p_A - \tau\vartheta, \qquad u_B(\vartheta) = \varrho - p_B - \tau(1 - \vartheta), \qquad u_0 = 0,$$

with an outside option normalized to zero.

If ecosystem i attracts app mass D_i, its training scope is

$$t_i \;=\; tD_i,$$

where t is the mass of covariates per app. We rule out multihoming by assumption and leave its analysis for future research.

Prediction value. Given training covariates t_i, the ecosystem chooses how many of the covariates to actually deploy in its prediction. Fixing user scale n, I suppress it from the endogenous objects' arguments, writing $V(t_i, t_0) = V(n, t_i, t_0)$ and $\mathcal{I}(t) = \mathcal{I}(n, t)$. Deployment of a target scope $t_0 \in [0, t_i]$ generates prediction value

$$\sigma_\beta^2 V(t_i, t_0) \;=\; \sigma_\beta^2 t_0 \mathcal{I}(t_i),$$

where $\sigma_\beta^2 > 0$ scales the prediction's value, $\xi \geq 0$ indexes the noise of the data-generating process, and the weight information $\mathcal{I}$ takes the closed form

$$\mathcal{I}(t) \;=\; \frac{n + 1 + \xi - \sqrt{(n + 1 + \xi)^2 - 4nt}}{2t},$$

is twice continuously differentiable, strictly increasing in t, strictly decreasing in ξ, and vanishes whenever n equals zero.

To deploy covariates as target covariates, the ecosystem must pay a cost to collect real-time data. I assume that deploying a mass of covariates t_0 results in a cost $C(t_0)$, strictly increasing

and strictly convex, with $C(0) = 0$ and $c(t_0) \equiv C'(t_0) > 0$ for all $t_0 \geq 0$.

The ecosystem's net data value, gross of app-fee revenues, is

$$\Phi(t_i) \equiv \max_{t_0 \in [0,t_i]} \left\{ \sigma_\beta^2 t_0 \mathcal{I}(t_i) - C(t_0) \right\}.$$

Ecosystem payoffs. Ecosystem i chooses f_i to maximize

$$\Pi_i = \Phi(tD_i) + \varrho f_i D_i,$$

where $\varrho f_i D_i$ is the fee revenue collected from apps.

Timing

Stage 1. Ecosystems A and B simultaneously choose their fees f_i to apps. Apps choose which ecosystem to join; demands D_i realize.

Stage 2. Each ecosystem observes its realized t_i and chooses its deployment scope $t_0^i \in [0, t_i]$. Prediction value $\Phi(t_i)$ realizes; payoffs clear.

I solve by backward induction and characterize the symmetric subgame-perfect equilibrium.

4.1 Stage 2: Deployment Choice

Fix any acquired data t_i. The deployment problem is

$$\Phi(t_i) = \max_{t_0 \in [0,t_i]} \left\{ \sigma_\beta^2 \, t_0 \, \mathcal{I}(t_i) \; - \; C(t_0) \right\}.$$

The marginal benefit of deploying covariate t_0 is $\sigma_\beta^2 \, \mathcal{I}(t_i)$, constant in t_0. The marginal cost is $c(t_0)$, strictly increasing. The interior optimum is $c^{-1}(\sigma_\beta^2 \mathcal{I}(t_i))$, strictly positive if and only if $\sigma_\beta^2 \mathcal{I}(t_i) > c(0)$.

Assumption 2. *For all $t \in [0, 1]$,*

$$c(0) < \sigma_\beta^2 \mathcal{I}(n, 1), \qquad \sigma_\beta^2 \mathcal{I}(n, t) < c(t).$$

The first inequality ensures that deployment is worthwhile once acquired scope is large enough. The second ensures that an ecosystem never wants to deploy all acquired covariates.

Lemma 1. *Under Assumption 2, the optimal deployment scope is*

$$\hat{t}_0(t) = \begin{cases} 0 & \text{if } t \leq \underline{t}, \\ c^{-1}\left(\sigma_\beta^2 \mathcal{I}(t)\right) & \text{if } t > \underline{t}, \end{cases}$$

where the participation threshold $\underline{t} \in (0, 1]$ *solves*

$$\sigma_\beta^2 \mathcal{I}\left(\underline{t}\right) = c(0).$$

Where interior,

$$\hat{t}_0'(t) \;=\; \sigma_\beta^2 \,\frac{\mathcal{I}'(t)}{c'(\hat{t}_0(t))} \;>\; 0,$$

and $\hat{t}_0$ *is strictly increasing in* σ_β^2 *and* n*, strictly decreasing in* ξ*.*

The threshold $\underline{t}$ has the reverse monotonicities: decreasing in n and σ_β^2, increasing in ξ. An ecosystem with insufficient scope generates zero data value, because the marginal weight estimate is too imprecise to overcome the sensitivity cost $c(0)$ of deploying even one covariate. Since ecosystem i acquires scope tD_i, the deployment threshold $\underline{t}$ induces a minimum profitable scope of apps

$$\underline{D} \equiv \frac{\underline{t}}{t}.$$

Thus

$$\Phi(tD_i) = 0 \quad \text{if } D_i \leq \underline{D}, \qquad \Phi(tD_i) > 0 \quad \text{if } D_i > \underline{D}.$$

4.2 The reduced-form value function

Substituting $\hat{t}_0$ into the deployment problem, the ecosystem's value as a function of acquired data is

$$\Phi(t) \;=\; \sigma_\beta^2 \,\hat{t}_0(t)\,\mathcal{I}(t) \;-\; C\left(\hat{t}_0(t)\right).$$

Corollary 3. Φ *is continuous and continuously differentiable on* $[0, 1]$*.* $\Phi(t) = 0$ *for* $t \leq \underline{t}$*, and* $\Phi(t) > 0$ *otherwise. On the positive region:*

(i)

$$\Phi'(t) = \sigma_\beta^2 \,\hat{t}_0(t)\,\mathcal{I}'(t) > 0.$$

(ii)

$$\Phi''(t) = \sigma_\beta^2 \Big[\underbrace{\hat{t}_0'(t) \cdot \mathcal{I}'(t)}_{t/t_0 \text{ complementarity: } > 0} + \underbrace{\hat{t}_0(t) \cdot \mathcal{I}''(t)}_{\text{IRS in } t\text{: } > 0} \Big] > 0.$$

4.3 Stage 1: Equilibrium

I characterize the interior (symmetric) equilibrium under a single global second-order assumption.

4.4 Demand and the symmetric profile

Assumption 3 (Full-coverage primitives)**.** *The app market is sufficiently valuable relative to horizontal differentiation:*

$$\varrho > 2\tau.$$

Under Assumption 3, whenever an ecosystem sets a fee $p_j \le \tau$, every app obtains $u_j(\vartheta) \ge \varrho - 2\tau > 0$ and strictly prefers participation. Since the equilibrium fee derived below satisfies $p^\star = \tau - t\Phi'(t/2) \le \tau$ (as $\Phi' \ge 0$), the outside option never binds at the symmetric profile or under a unilateral deviation, and the standard full-coverage Hotelling demand system applies. The app indifferent between A and B solves

$$\varrho - p_A - \tau\hat{\vartheta} = \varrho - p_B - \tau(1 - \hat{\vartheta}).$$

Therefore

$$\hat{\vartheta} = \frac{1}{2} + \frac{p_B - p_A}{2\tau}, \qquad D_A = \hat{\vartheta}, \qquad D_B = 1 - \hat{\vartheta}, \tag{3}$$

with an interior allocation whenever $|p_A - p_B| < \tau$.

Ecosystem i's payoff is

$$\Pi_i(p_i; p_j) = \Phi\big(tD_i(p_i, p_j)\big) + p_i D_i(p_i, p_j), \tag{4}$$

where

$$\frac{\partial D_i}{\partial p_i} = -\frac{1}{2\tau}.$$

The first-order condition is

$$\frac{\partial \Pi_i}{\partial p_i} = -\frac{t}{2\tau}\Phi'(tD_i) + D_i - \frac{p_i}{2\tau} = 0, \tag{5}$$

and the second-order condition is

$$\frac{\partial^2 \Pi_i}{\partial p_i^2} = \frac{t^2}{4\tau^2}\Phi''(tD_i) - \frac{1}{\tau} < 0. \tag{6}$$

I impose this condition uniformly:

Assumption 4 (Global SOC)**.** $t^2\,\Phi''(\tilde{t}) \;<\; 4\tau$ *for all* $\tilde{t} \in [0, 1]$.

Assumption 4 is the right global condition for two reasons.

First, it implies $\Pi_i(p_i; p_j)$ is strictly concave in p_i over the range of demand $D_i \in [\underline{D}, 1]$ on which Φ is positive. Where $\Phi = 0$, concavity is immediate: there $\Pi_i = p_i D_i$ with D_i affine and strictly decreasing in p_i, so Π_i is a strictly concave quadratic with $\partial^2\Pi_i/\partial p_i^2 = -1/\tau < 0$. Second, it is the cleanest version of the local SOC that guarantees no *single-margin* non-local deviation can beat the symmetric profile.

Assumption 5 (Positive data value at symmetry)**.**

$$\underline{D} < \frac{1}{2}, \quad \textit{or equivalently,} \quad \underline{t} < \frac{t}{2}.$$

This assumption guarantees that the symmetric allocation lies in the positive-Φ region.

4.5 Interior equilibrium

Lemma 2 (Interior equilibrium)**.** *Under Assumptions 3, 4, and 5, the unique symmetric pure-strategy equilibrium has*

$$p^\star \;=\; \tau - t\,\Phi'(t/2), \qquad f^\star \;=\; \frac{\tau - t\,\Phi'(t/2)}{\varrho},$$

demand $D_i^\star = 1/2$*, scope* $t_i^\star = t/2$*, and profit*

$$\Pi_i^\star \;=\; \Phi(t/2) \;+\; \tfrac{1}{2}\big[\tau - t\Phi'(t/2)\big] \;=\; \Phi(t/2) \;-\; \tfrac{t}{2}\Phi'(t/2) \;+\; \tfrac{\tau}{2}.$$

Moreover, $p^\star$ *is a Nash equilibrium against* any *unilateral deviation, including non-local ones.*

Proof. Existence and the formula follow from (5) evaluated at $D_i = 1/2$. Under Assumption 4, $\Pi_i(\cdot; p^\star)$ is strictly concave on the demand range where $\Phi > 0$. Where $\Phi = 0$ (i.e., $D_i \le \underline{D}$), $\Pi_i(p_i; p^\star) = p_i D_i$ is a concave quadratic in p_i (since D_i is linear in $-p_i$).

At the boundary $D_i = \underline{D}$, the two payoff pieces meet continuously and have matching first derivatives. Indeed, $\Phi = 0$ below the deployment threshold $\underline{t}$, so the left derivative of Φ at $\underline{t}$ is zero. Above the threshold, the envelope theorem gives

$$\Phi'(s) = \sigma_\beta^2 \hat{t}_0(s) \mathcal{I}'(s).$$

Since $\hat{t}_0(\underline{t}) = 0$, the right derivative also satisfies

$$\Phi'(\underline{t}^+) = 0 = \Phi'(\underline{t}^-).$$

Hence Φ is C^1 at the threshold. It follows that $\Pi_i(\cdot; p^\star)$ is concave on the entire interval, and $p^\star$ is the unique global best response.

For prices inducing $D_i = 1$, profit equals $\Phi(t) + p_i$ and is maximized at the boundary price $p_i = p^\star - \tau$. For prices inducing $D_i = 0$, profit is zero. Both boundary payoffs are strictly below the payoff at $p^\star$ by strict concavity on $[p^\star - \tau, p^\star + \tau]$.

□

The economic content of $p^\star$ has a simple interpretation: when $\tau < t\Phi'(t/2)$ the data feedback dominates and ecosystems subsidize apps ($p^\star < 0$, negative ad valorem); when $\tau > t\Phi'(t/2)$, the differentiation rent dominates and ecosystems charge a positive fee.

4.6 Strong feedback and tipping pressure

The global second-order condition in Assumption 4 rules out strong data feedback. Its failure has a natural interpretation: the symmetric allocation is no longer locally stable. In this sense, economies of scope can generate tipping pressure even though the proposition above focuses on the symmetric equilibrium.

Proposition 6 (Local tipping pressure)**.** *Suppose the app market is fully covered and the symmetric allocation lies in the positive-value region. If*

$$t^2\Phi''(t/2) > 4\tau,$$

then the symmetric price candidate

$$p^\star = \tau - t\Phi'(t/2)$$

is not a local Nash equilibrium. In particular, for all sufficiently small $\varepsilon > 0$, a deviation to $p_i = p^\star - \varepsilon$, which attracts more apps, is profitable.

Proof. At the symmetric candidate, the first-order condition holds:

$$\left.\frac{\partial \Pi_i}{\partial p_i}\right|_{p_i = p^\star} = 0.$$

The second derivative at the symmetric allocation is

$$\left.\frac{\partial^2 \Pi_i}{\partial p_i^2}\right|_{p_i = p^\star} = \frac{t^2}{4\tau^2}\Phi''(t/2) - \frac{1}{\tau}.$$

Thus $t^2\Phi''(t/2) > 4\tau$ implies

$$\left.\frac{\partial^2 \Pi_i}{\partial p_i^2}\right|_{p_i = p^\star} > 0.$$

A Taylor expansion gives

$$\Pi_i(p^\star - \varepsilon; p^\star) - \Pi_i(p^\star; p^\star) = \frac{1}{2}\left[\frac{t^2}{4\tau^2}\Phi''(t/2) - \frac{1}{\tau}\right]\varepsilon^2 + o(\varepsilon^2).$$

For all sufficiently small $\varepsilon > 0$, this expression is positive. The deviation lowers the fee and increases app demand above $1/2$. Hence the symmetric candidate is locally unstable. □

Tipping pressure is stronger when horizontal differentiation among apps is weak (τ is low), each app contributes many covariates (t is high), prediction is valuable (σ_β^2 is high), data are precise (ξ is low), the ecosystem has sufficient user scale (n is high), and deploying additional covariates is cheap ($c(0)$ and $c'(\cdot)$ are low).

This mechanism speaks directly to current policy debates on digital ecosystems. Rules on data combination, consent, interoperability, and gatekeeper conduct affect the primitives that determine tipping pressure. Restrictions on cross-app or cross-service data combination reduce the effective scope contributed by each app (t) or raise the cost of deploying covariates ($c(0)$ and $c'(\cdot)$), thereby weakening the feedback from app adoption to prediction quality. By contrast, when ecosystems can freely combine rich app data, when prediction is central to monetization (σ_β^2 is high), and when app developers can be attracted by small fee cuts or subsidies (τ is low), the model predicts stronger tipping pressure. The policy concern is therefore not only whether a platform has many users, but whether user scale, app scope, and data combination reinforce one another strongly enough to make symmetric competition unstable.

Corollary 4 (No symmetric equilibrium under strong feedback)**.** *Under the hypotheses of Proposition 6, the symmetric profile $p^\star = \tau - t\Phi'(t/2)$ is the unique symmetric candidate satisfying the first-order condition, and it fails the second-order condition. Hence no symmetric price profile is a (local) Nash equilibrium: if a pure-strategy equilibrium exists, it is asymmetric.*

Proof. A symmetric profile has $D_i = 1/2$; the first-order condition (5) then pins down $p^\star$ uniquely. Any other symmetric profile violates (5) and is not a best response. At $p^\star$, Proposition 6 shows $\partial^2\Pi_i/\partial p_i^2 > 0$, so $p^\star$ is not a local best response either. Thus no symmetric profile is a local equilibrium. □

Remark 1. Proposition 6 and Corollary 4 establish that strong data feedback destabilizes symmetric competition and that there exists no symmetric equilibrium in pure strategies. They do *not* establish existence of a tipped (asymmetric or corner) equilibrium, which would require analyzing corner allocations $D_i \in \{0, 1\}$ and is left for future work. I therefore speak of tipping *pressure* rather than of a tipped equilibrium.

5 Conclusion

This paper develops a theory of the value of data for prediction. The central message is simple: data are not homogeneous. Their value depends on the interaction of three margins: the number of training observations, the covariates observed in the training sample, and the covariates observed at prediction time. I decompose dataset value into two components. The target signal determines the individual attributes to which learning can be applied. Weight information measures how much the data resolve prior uncertainty about the mapping from covariates to outcomes. Target covariates supply the first component; training data supply the second.

Two results follow. First, training covariates exhibit economies of scope. An additional covariate is more valuable when the firm already observes many covariates, because broader

scope reduces residual noise and sharpens learning about all trained weights. Second, observations and covariates are complements in small datasets but can become substitutes in large ones. These results show why diminishing returns to observations — the classic law-of-large-numbers intuition — do not imply diminishing returns to data as a whole. Even when additional observations have declining value, returns to data can remain increasing through economies of scope in parameter learning.

These statistical forces shape market structure. In the application, two ecosystems compete for apps, which supply covariates, while each ecosystem holds a captive mass of users, who supply observations. Economies of scope create a feedback loop: attracting more apps raises the value of an ecosystem's data, which increases its willingness to compete for still more apps. When prediction value is low, the data-generating process is noisy, or apps are sufficiently differentiated, this loop is too weak to destabilize symmetric competition. When prediction value is high and app data are rich, the same loop can make the symmetric allocation locally unstable. A small fee cut attracts more apps, expands scope, and raises the return to attracting still more. Whether the market tips therefore depends on whether the data-feedback force is strong enough to overcome horizontal differentiation.

For competition policy, this qualifies both sides of the debate. Concentration can arise from the statistical structure of prediction itself even in the absence of demand-side network effects or exogenous entry barriers. The U.S. House Report's concern that data can serve as a barrier to entry is therefore conditionally justified. The concern is strongest when data are rich, predictions are valuable and precise, and the user base is large enough to sustain feedback but not so large that additional observations have exhausted their learning value. Outside that regime, symmetric competition can survive even with economies of scope. Policies that change the cost of deploying covariates, the substitutability between ecosystems, or firms' ability to combine covariate sources can therefore move a market across the tipping boundary.

The last lever — combining covariate sources — matters directly for merger review, where the model suggests that the key indicator is the dimension across which data are merged. Merging the same covariates across more users runs into diminishing returns to scale: the merged dataset is worth strictly less than the two standalone datasets, weakening the efficiency defense. Merging different covariates on the same users generates additional value thanks to the economies of scope, so the merging firms can build an efficiency defense. The same forces shape acquisition incentives. Covariate-rich entrants are most valuable to scale-rich incumbents precisely when observations and covariates are complements, generating a rich-get-richer acquisition motive that may fade once the incumbent has acquired abundant data, especially if data are not noisy.

Several extensions are natural. The model's empirical content can be tested by estimating how predictive performance varies separately with training scale, training scope, and target scope. Moreover, the model can serve as a preliminary step to develop a tool to assign prices to data based on the statistical value that an algorithm can extract from them. Finally, embed-

ding the value function into dynamic models of platform competition with data sharing, and multihoming could provide evaluation on whether existing policies to promote competition suppress the economies of scope that make data valuable.

References

Acemoğlu, D., A. Makhdoumi, A. Malekian, and A. Özdağlar (2022). "Too much data: Prices and inefficiencies in data markets." In: *American Economic Journal: Microeconomics* 14.4, pp. 218–256.

Aral, S., E. Brynjolfsson, and D. Wu (2008). "Which came first, IT or productivity? The virtuous cycle of investment and use in enterprise systems." In.

Bai, Z. and J. Silverstein (2009). *Spectral Analysis of Large Dimensional Random Matrices*. Springer Series in Statistics. Springer New York.

Bajari, P., V. Chernozhukov, A. Hortaçsu, and J. Suzuki (2019). "The impact of big data on firm performance: An empirical investigation." In: *AEA papers and proceedings* 109, pp. 33–37.

Bergemann, D. and A. Bonatti (2024). "Data, Competition, and Digital Platforms." In: *American Economic Review* 114.8, pp. 2553–2595.

Bergemann, D., A. Bonatti, and T. Gan (2022). "The economics of social data." In: *The RAND Journal of Economics* 53.2, pp. 263–296.

Bergemann, D., A. Bonatti, and A. Smolin (2018). "The Design and Price of Information." In: *American Economic Review* 108.1, pp. 1–48.

Calzolari, G., A. Cheysson, and R. Rovatti (2025). "Machine data: market and analytics." In: *Management Science*.

Carballa-Smichowski, B., N. Duch-Brown, S. Höcük, P. Kumar, B. Martens, J. Mulder, and P. Prüfer (2025a). "Economies of scope in data aggregation: Evidence from health data." In: *Information Economics and Policy* 71, p. 101146.

Carballa-Smichowski, B., Y. Lefouili, A. Mantovani, and C. Reggiani (2025b). *Data Sharing or Analytics Sharing?* TSE Working Paper 25-1615. Toulouse School of Economics.

Cornière, A. de and G. Taylor (2024). "Data-Driven Mergers." In: *Management Science* 70.9, pp. 6473–6482.

Dasaratha, K., J. Ortner, and C. Zhu (2025). "Markets for Models." In: *arXiv preprint arXiv:2503.02946*.

De Corniere, A. and G. Taylor (2025). "Data and Competition: A Simple Framework." In: *Forthcoming, RAND Journal of Economics*.

Digital Platforms, S. C. on (2019). *Final Report*. Stigler Center for the Study of the Economy and the State, University of Chicago Booth School of Business.

Farboodi, M. and L. Veldkamp (2025). *A model of the Data Economy*. Tech. rep. R&R, Review of Economic Studies.

Goldfarb, A. and C. Tucker (2011). "Privacy Regulation and Online Advertising." In: *Management Science* 57.1, pp. 57–71.

Gu, Y., L. Madio, and C. Reggiani (2021). "Data brokers co-opetition." In: *Oxford Economic Papers* 74.3, pp. 820–839.

Hagiu, A. and J. Wright (2023). "Data-enabled learning, network effects, and competitive advantage." In: *The RAND Journal of Economics* 54.4, pp. 638–667.

Hastie, T., A. Montanari, S. Rosset, and R. J. Tibshirani (2020). "Surprises in High-Dimensional Ridgeless Least Squares Interpolation." In.

Iyer, G. and T. Ke (2024). "Competition and Algorithmic Complexity in Predictive Analytics." In: *Marketing Science* 43.2, pp. 215–233.

Jones, C. I. and C. Tonetti (2020). "Nonrivalry and the Economics of Data." In: *American Economic Review* 110.9, pp. 2819–58.

Lee, G. and J. Wright (2023). "Recommender systems and the Value of User Data." In: *National University of Singapore Working Paper*.

Marčenko, V. A. and L. A. Pastur (1967). "Distribution of eigenvalues for some sets of random matrices." In: *Mathematics of the USSR-Sbornik* 1.4, pp. 457–483.

Montiel Olea, J. L., P. Ortoleva, M. M. Pai, and A. Prat (2022). "Competing Models." In: *The Quarterly Journal of Economics* 137.4, pp. 2419–2457.

Prüfer, J. and C. Schottmüller (2021). "Competing with big data." In: *The Journal of Industrial Economics* 69.4, pp. 967–1008.

Rhodes, A., J. Zhou, and J. Zhou (2026). *Digital Ecosystems and Data Regulation*. TSE Working Paper 25-1621. Revised February 2026. Toulouse School of Economics.

Schaefer, M. (2025). *When Should we Expect Non-Decreasing Returns from Data in Prediction Tasks?*

Schaefer, M. and G. Sapi (2023). "Complementarities in learning from data: Insights from general search." In: *Information Economics and Policy* 65, p. 101063.

Schmidt, E. (2009). *How Google Plans to Stay Ahead in Search*. Accessed: 2025-11-12. Bloomberg. URL: https://www.bloomberg.com/news/articles/2009-10-02/how-google-plans-to-stay-ahead-in-search.

Strzalecki, T. (2024). *Variational Bayes and non-Bayesian Updating*.

UK Competition and Markets Authority (2019). *Unlocking Digital Competition: Report of the Digital Competition Expert Panel*. London: Competition and Markets Authority.

Varian, H. (2015). "Is there a Data Barrier to Entry?" In: *Accessed at https://www.learconference2015.com/wp-content/uploads/2014/11/Varian-slides.pdf in March 2025.*

Varian, H. (2018). "Artificial Intelligence, Economics, and Industrial Organization." In: *The Economics of Artificial Intelligence: An Agenda*. NBER Chapters. National Bureau of Economic Research, Inc, pp. 399–419.

A Proofs

The whole appendix rests on one structural observation. Split the target of a training individual into observed-and-trained covariates and the rest:

$$y_i = \boldsymbol{x}_{i,\mathcal{T}}^\top \boldsymbol{\beta}_{\mathcal{T}} + w_i, \qquad w_i \equiv \sum_{k \notin \mathcal{T}} \beta_k x_{ik} + \eta_i.$$

The residual w_i is *not* Gaussian. But conditional on the single scalar

$$s^2 \;\equiv\; \sigma_\eta^2 \sigma_\beta^2 + \frac{\sigma_x^2}{K} \sum_{k \notin \mathcal{T}} \beta_k^2 \;=\; \sigma_\eta^2 \sigma_\beta^2 + \frac{\sigma_x^2}{K} \|\boldsymbol{\beta}_{\mathcal{T}^c}\|^2,$$

it *is* exactly Gaussian, and s^2 concentrates. Conditioning on s^2 therefore turns the model into an exact Gaussian linear model whose noise level is known up to a vanishing error. This is Lemma 3, from which the optimal predictor, the closed form, and the value formula all follow by elementary steps. Throughout, $A_K \equiv \boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{X}_{N,\mathcal{T}}$, $\sigma_\varepsilon^2(t) \equiv (1-t)\sigma_x^2 + \sigma_\eta^2$, and $\bar{s}^2 \equiv \sigma_\beta^2 \sigma_\varepsilon^2(t)$; I maintain $\sigma_\varepsilon^2(t) > 0$ (automatic unless $t = 1$ and $\sigma_\eta^2 = 0$).

A.1 The scalar-noise reduction

Lemma 3 (Scalar-noise reduction)**.** *Fix K and a design $(N, \mathcal{T}, \mathcal{T}_0)$. Conditional on $(\boldsymbol{X}_{N,\mathcal{T}}, s^2)$:*

1. *$\boldsymbol{\beta}_{\mathcal{T}} \sim \mathcal{N}(0, \sigma_\beta^2 I_T)$ and $\boldsymbol{w}_N \sim \mathcal{N}(0, s^2 I_N)$ are independent, and $\boldsymbol{y}_N = \boldsymbol{X}_{N,\mathcal{T}} \boldsymbol{\beta}_{\mathcal{T}} + \boldsymbol{w}_N$; consequently the posterior of $\boldsymbol{\beta}_{\mathcal{T}}$ is Gaussian,*

$$\boldsymbol{\beta}_{\mathcal{T}} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}, s^2 \;\sim\; \mathcal{N}\big(\boldsymbol{\mu}(s^2),\, \Sigma(s^2)\big), \qquad \Sigma(s^2) = \sigma_\beta^2 \Big(I_T + \tfrac{\sigma_\beta^2}{s^2} A_K \Big)^{-1}, \quad \boldsymbol{\mu}(s^2) = \big(\tfrac{s^2}{\sigma_\beta^2} I_T + A_K \big)^{-1} \boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{y}_N.$$

2. *As $K \to \infty$, $s^2 \xrightarrow{p} \bar{s}^2$, with $\mathbb{E} s^2 \to \bar{s}^2$ and $\mathrm{Var}(s^2) = O(1/K)$.*

Proof. Part 1. Write $\boldsymbol{w}_N = \boldsymbol{X}_{N,\mathcal{T}^c} \boldsymbol{\beta}_{\mathcal{T}^c} + \boldsymbol{\eta}_N$. Given $\boldsymbol{\beta}_{\mathcal{T}^c}$, the N coordinates $w_i = \sum_{k \notin \mathcal{T}} \beta_k x_{ik} + \eta_i$ are independent linear combinations of the independent Gaussians $\{x_{ik}\}_{k \notin \mathcal{T}}$ and η_i; hence $\boldsymbol{w}_N \mid \boldsymbol{\beta}_{\mathcal{T}^c} \sim \mathcal{N}(0, s^2 I_N)$, a law that depends on $\boldsymbol{\beta}_{\mathcal{T}^c}$ *only through* s^2. Averaging over $\{\boldsymbol{\beta}_{\mathcal{T}^c} : s^2(\boldsymbol{\beta}_{\mathcal{T}^c}) = s^2\}$ leaves this law unchanged, so $\boldsymbol{w}_N \mid s^2 \sim \mathcal{N}(0, s^2 I_N)$. Because $\boldsymbol{\beta}_{\mathcal{T}}$ is independent of $(\boldsymbol{X}_{N,\mathcal{T}^c}, \boldsymbol{\beta}_{\mathcal{T}^c}, \boldsymbol{\eta}_N)$ and of $\boldsymbol{X}_{N,\mathcal{T}}$, it is independent of $\boldsymbol{w}_N$ given $(\boldsymbol{X}_{N,\mathcal{T}}, s^2)$ and retains its prior $\mathcal{N}(0, \sigma_\beta^2 I_T)$. This is a Gaussian linear model; the stated posterior is the standard conjugate formula, with precision $\sigma_\beta^{-2} I_T + s^{-2} A_K$.

Part 2. The summands β_k^2 $(k \notin \mathcal{T})$ are i.i.d. with mean σ_β^2 and variance $2\sigma_\beta^4$, and $|\mathcal{T}^c| = K - T$ with $T/K \to t$. Hence $\mathbb{E} s^2 = \sigma_\eta^2 \sigma_\beta^2 + \sigma_x^2 \sigma_\beta^2 \frac{K-T}{K} \to \bar{s}^2$ and $\mathrm{Var}(s^2) = \big(\frac{\sigma_x^2}{K}\big)^2 (K-T)\, 2\sigma_\beta^4 = O(1/K)$, so $s^2 \xrightarrow{p} \bar{s}^2$ by Chebyshev. □

A.2 Optimal predictor (Proposition 1)

Proof. Under quadratic loss the posterior mean minimizes posterior risk, so $\hat{y}_D^*(\boldsymbol{d}) = \mathbb{E}[y_0 \mid \boldsymbol{d}]$. Writing $y_0 = \sum_k \beta_k x_{0k} + \eta_0$ and using $\mathbb{E}[\eta_0 \mid \boldsymbol{d}] = 0$, take a coordinate k. If $k \notin \mathcal{T}_0$, then x_{0k} is unobserved, centered, and independent of $(\boldsymbol{d}, \beta_k)$, so $\mathbb{E}[\beta_k x_{0k} \mid \boldsymbol{d}] = 0$. If $k \in \mathcal{T}_0$, then x_{0k} is observed and independent of $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{\beta})$, so $\mathbb{E}[\beta_k x_{0k} \mid \boldsymbol{d}] = x_{0k}\,\mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}]$. Finally, for $k \notin \mathcal{T}$ the conditional likelihood of $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ is even in β_k (only $\beta_k x_{ik}$ involves β_k, and x_{ik} is unobserved, centered, symmetric), and the prior is symmetric, so the posterior of β_k is symmetric and $\mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}] = 0$. Only $k \in \mathcal{T}_0 \cap \mathcal{T}$ survive:

$$\hat{y}_D^*(\boldsymbol{d}) = \boldsymbol{x}_{0,\mathcal{T}_0\cap\mathcal{T}}^\top \mathbb{E}[\boldsymbol{\beta}_{\mathcal{T}_0\cap\mathcal{T}} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}]. \qquad \square$$

For nested designs ($\mathcal{T}_0 \subseteq \mathcal{T}$) this reads $\hat{y}^* = \boldsymbol{x}_{0,\mathcal{T}_0}^\top \boldsymbol{\mu}_{\mathcal{T}_0}$ with $\boldsymbol{\mu} \equiv \mathbb{E}[\boldsymbol{\beta}_{\mathcal{T}} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}]$.

A.3 Posterior-trace equivalence

This is the step invoked in the main text. It says the average posterior variance of the trained weights is asymptotically the ridge value at the *effective* noise $\bar{s}^2$, despite the true residual being non-Gaussian.

Lemma 4 (Posterior-trace equivalence)**.** *Under Assumption 1,*

$$\frac{1}{T}\operatorname{tr}\operatorname{Var}\!\big(\boldsymbol{\beta}_{\mathcal{T}} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}\big) \;-\; \frac{1}{T}\operatorname{tr}\,\Sigma(\bar{s}^2) \;\xrightarrow{p}\; 0, \qquad \Sigma(\bar{s}^2) = \sigma_\beta^2\Big(I_T + \tfrac{1}{\sigma_\varepsilon^2(t)} A_K\Big)^{-1}.$$

Proof. Condition on s^2 and apply the law of total variance over its posterior:

$$\operatorname{Var}(\boldsymbol{\beta}_{\mathcal{T}} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}) = \underbrace{\mathbb{E}\big[\Sigma(s^2) \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}\big]}_{\text{within}} + \underbrace{\operatorname{Var}\big(\boldsymbol{\mu}(s^2) \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}\big)}_{\text{between}},$$

using Lemma 3(1) for the conditional mean $\boldsymbol{\mu}(s^2)$ and covariance $\Sigma(s^2)$. We show the normalized trace of the between term is $o_p(1)$ and that the within term converges to $\frac{1}{T}\operatorname{tr}\Sigma(\bar{s}^2)$.

Between term. Write $\boldsymbol{\mu}(s^2) = (cI_T + A_K)^{-1}\boldsymbol{b}$ with $c \equiv s^2/\sigma_\beta^2$ and $\boldsymbol{b} \equiv \boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{y}_N$. The resolvent identity gives, for any c_1, c_2,

$$\boldsymbol{\mu}(c_1) - \boldsymbol{\mu}(c_2) = (c_2 - c_1)(c_1 I + A_K)^{-1}(c_2 I + A_K)^{-1}\boldsymbol{b}, \qquad \|\boldsymbol{\mu}(c_1) - \boldsymbol{\mu}(c_2)\| \le \frac{|c_1 - c_2|}{c_1 c_2}\|\boldsymbol{b}\|,$$

since $A_K \succeq 0$ implies $\|(cI + A_K)^{-1}\|_{\mathrm{op}} \le c^{-1}$. Because $c \xrightarrow{p} \sigma_\varepsilon^2(t) > 0$, fix $\underline{c} \in (0, \sigma_\varepsilon^2(t))$; on the event $\{c \ge \underline{c}\}$, whose probability $\to 1$, drawing c, c' independently from the posterior of s^2/σ_β^2,

$$\frac{1}{T}\operatorname{tr}\operatorname{Var}(\boldsymbol{\mu}(s^2) \mid \cdot) = \frac{1}{2T}\,\mathbb{E}\big[\|\boldsymbol{\mu}(c) - \boldsymbol{\mu}(c')\|^2 \mid \cdot\big] \le \frac{1}{\underline{c}^4}\operatorname{Var}(c \mid \cdot)\,\frac{\|\boldsymbol{b}\|^2}{T}.$$

Two bounds finish this term. First, by the law of total variance, $\mathbb{E}[\mathrm{Var}(c \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})] \le \mathrm{Var}(c) = \sigma_\beta^{-4}\,\mathrm{Var}(s^2) = O(1/K)$, so $\mathrm{Var}(c \mid \cdot) = O_p(1/K)$. Second, $\mathbb{E}\|\boldsymbol{b}\|^2 = \mathbb{E}\|\boldsymbol{X}_{N,\mathcal{T}}^\top \boldsymbol{y}_N\|^2 = O(K)$: each of the T entries $\sum_i x_{ik} y_i$ has second moment $O(N/K)$ (the diagonal terms $\mathbb{E}[x_{ik}^2 y_i^2] = O(1/K)$ dominate, being N in number), and $T = O(K)$, $N = O(K)$. Hence $\|\boldsymbol{b}\|^2/T = O_p(1)$ and the between trace is $O_p(1/K)$.

Within term. Let $g(s^2) \equiv \frac{1}{T}\,\mathrm{tr}\,\Sigma(s^2) = \frac{\sigma_\beta^2}{T}\,\mathrm{tr}[(I_T + \frac{\sigma_\beta^2}{s^2} A_K)^{-1}] \in [0, \sigma_\beta^2]$. It is increasing in s^2 (as $s^2 \uparrow$, $\frac{\sigma_\beta^2}{s^2} A_K \downarrow$ in the Loewner order, so $(I + \cdot)^{-1} \uparrow$), and its derivative obeys

$$g'(s^2) = \frac{\sigma_\beta^4}{s^4 T}\,\mathrm{tr}\Big[A_K \big(I_T + \tfrac{\sigma_\beta^2}{s^2} A_K\big)^{-2}\Big] \le \frac{\sigma_\beta^4}{s^4}\,\frac{\mathrm{tr}\,A_K}{T} = \frac{\sigma_\beta^4}{s^4}\,\mathcal{S}_K,$$

using $A_K (I + \frac{\sigma_\beta^2}{s^2} A_K)^{-2} \preceq A_K$.

It remains to control

$$\mathcal{S}_K \equiv \frac{1}{T}\,\mathrm{tr}\,A_K = \frac{1}{T} \sum_{k \in \mathcal{T}} \sum_{i=1}^{N} x_{ik}^2.$$

Since the x_{ik}'s are independent with

$$\mathbb{E}[x_{ik}^2] = \frac{\sigma_x^2}{K}, \qquad \mathrm{Var}(x_{ik}^2) = \frac{2\sigma_x^4}{K^2},$$

I have

$$\mathbb{E}[\mathcal{S}_K] = \frac{N}{K}\sigma_x^2 \to n\sigma_x^2$$

and

$$\mathrm{Var}(\mathcal{S}_K) = \frac{1}{T^2} N T \frac{2\sigma_x^4}{K^2} = \frac{2N\sigma_x^4}{TK^2} \to 0.$$

Hence

$$\mathcal{S}_K \xrightarrow{p} n\sigma_x^2.$$

On $\{s^2 \ge \sigma_\beta^2 \underline{c}\}$ and since $\mathcal{S}_K \xrightarrow{p} n\sigma_x^2$, g is Lipschitz with an $O_p(1)$ constant L. Since the within term equals $\mathbb{E}[g(s^2) \mid \cdot]$,

$$\big|\mathbb{E}[g(s^2) \mid \cdot] - g(\bar{s}^2)\big| \le L\,\mathbb{E}\big[\,|s^2 - \bar{s}^2| \mid \cdot\,\big], \qquad \mathbb{E}\Big(\mathbb{E}[\,|s^2 - \bar{s}^2| \mid \cdot\,]\Big) = \mathbb{E}|s^2 - \bar{s}^2| \to 0$$

by Lemma 3(2) (L^1 convergence follows from $\mathbb{E}s^2 \to \bar{s}^2$ and $\mathrm{Var}(s^2) \to 0$). Thus $\mathbb{E}[\,|s^2 - \bar{s}^2| \mid \cdot\,] \xrightarrow{p} 0$ and the within term converges to $g(\bar{s}^2) = \frac{1}{T}\,\mathrm{tr}\,\Sigma(\bar{s}^2)$. Combining the two terms proves the claim. □

A.4 Closed form for weight information (Proposition 7)

By definition $\mathcal{I}(n,t) = 1 - \lim_K \frac{1}{T\sigma_\beta^2}\,\mathrm{tr}\,\mathrm{Var}(\boldsymbol{\beta}_\mathcal{T} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$, which by Lemma 4 equals $\lim_K \frac{1}{T}\,\mathrm{tr}\big[I_T - (I_T + \frac{1}{\sigma_\varepsilon^2} A_K)^{-1}\big] = \lim_K \frac{1}{T}\,\mathrm{tr}\big[A_K (A_K + \sigma_\varepsilon^2 I_T)^{-1}\big]$.

Proposition 7 (Closed form for weight information). *Under Assumption 1, with* $\xi \equiv \sigma_\eta^2/\sigma_x^2$,

$$\mathcal{I}(n,t) = \frac{n+1+\xi-\sqrt{(n+1+\xi)^2-4nt}}{2t} = \frac{2n}{n+1+\xi+\sqrt{(n+1+\xi)^2-4nt}}.$$

Proof. Write $\boldsymbol{X}_{N,T} = \sqrt{\sigma_x^2/K}\,\boldsymbol{Z}_K$ with $\boldsymbol{Z}_K$ having i.i.d. standard normal entries, so $A_K = \frac{\sigma_x^2 N}{K} S_K$ with $S_K \equiv N^{-1}\boldsymbol{Z}_K^\top \boldsymbol{Z}_K$. Then

$$\frac{1}{T}\operatorname{tr}\left[A_K(A_K+\sigma_\varepsilon^2 I_T)^{-1}\right] = \frac{1}{T}\operatorname{tr}\left[S_K(S_K+\kappa_K I_T)^{-1}\right], \qquad \kappa_K = \frac{\sigma_\varepsilon^2}{\sigma_x^2}\frac{K}{N} \to \kappa \equiv \frac{\sigma_\varepsilon^2(t)}{n\sigma_x^2} = \frac{1-t+\xi}{n}.$$

The empirical spectrum of S_K converges to the Marchenko–Pastur law μ_γ with aspect ratio $\gamma_K = T/N \to \gamma \equiv t/n$ (Marčenko and Pastur 1967; Bai and Silverstein 2009). Since $x \mapsto x/(x+\kappa)$ is bounded and continuous on $[0,\infty)$,

$$\mathcal{I}(n,t) = \int \frac{x}{x+\kappa}\, d\mu_\gamma(x) = 1-\kappa\, m_\gamma(-\kappa), \qquad m_\gamma(z) = \frac{1-\gamma-z-\sqrt{(z-a_\gamma)(z-b_\gamma)}}{2\gamma z},$$

the Stieltjes transform of μ_γ, with $a_\gamma = (1-\sqrt{\gamma})^2$, $b_\gamma = (1+\sqrt{\gamma})^2$. Evaluating at $z=-\kappa$,

$$\mathcal{I}(n,t) = \frac{1+\gamma+\kappa-\sqrt{(\kappa+a_\gamma)(\kappa+b_\gamma)}}{2\gamma}.$$

Substituting $\gamma = t/n$ and $\kappa = (1-t+\xi)/n$ gives $1+\gamma+\kappa = \frac{1}{n}(n+1+\xi)$ and $\kappa + a_\gamma = \frac{1}{n}(n+1+\xi-2\sqrt{nt})$, $\kappa+b_\gamma = \frac{1}{n}(n+1+\xi+2\sqrt{nt})$, so $\sqrt{(\kappa+a_\gamma)(\kappa+b_\gamma)} = \frac{1}{n}\sqrt{(n+1+\xi)^2-4nt}$. The claim follows; the second expression is the rationalized form. □

Equivalently, $\mathcal{I} = \mathcal{I}(n,t)$ is the root in $[0,1]$ of the quadratic

$$t\,\mathcal{I}^2 - (n+1+\xi)\,\mathcal{I} + n = 0 \qquad \text{(normalizing } \sigma_x^2 = 1\text{)}, \tag{7}$$

namely the smaller root, which is the one lying in $[0,1]$.

A.5 Self-averaging and subset concentration

Lemma 5 (Self-averaging of posterior second moments). *For any deterministic* $\mathcal{A}_K \subseteq \{1,\dots,K\}$ *and any* σ*-field* $\mathcal{F}_K$ *generated by the observed dataset,*

$$\frac{1}{K}\sum_{k\in\mathcal{A}_K}\left(\mathbb{E}[\beta_k^2 \mid \mathcal{F}_K] - \sigma_\beta^2\right) \xrightarrow{p} 0.$$

Proof. Let $Z_K = \frac{1}{K}\sum_{k\in\mathcal{A}_K}(\beta_k^2-\sigma_\beta^2)$. The displayed average is $\mathbb{E}[Z_K \mid \mathcal{F}_K]$, and by conditional Jensen $\mathbb{E}[(\mathbb{E}[Z_K \mid \mathcal{F}_K])^2] \le \mathbb{E}[Z_K^2] = \frac{|\mathcal{A}_K|}{K^2}\,2\sigma_\beta^4 \le \frac{2\sigma_\beta^4}{K} \to 0$. Hence the average $\to 0$ in L^2, so in probability. □

Lemma 6 (Subset concentration). *Let $(a_k)_{k\in\mathcal{T}(K)}$ be exchangeable with $\sup_k \mathbb{E}a_k^2 < \infty$, and suppose the full-block average $\frac{1}{T}\sum_{k\in\mathcal{T}} a_k \xrightarrow{p} L$. Then for any deterministic $\mathcal{T}_0(K) \subseteq \mathcal{T}(K)$ with $|\mathcal{T}_0(K)| \to \infty$, $\frac{1}{T_0}\sum_{k\in\mathcal{T}_0} a_k \xrightarrow{p} L$.*

Proof. By exchangeability $\mathbb{E}a_k = \mathbb{E}[\frac{1}{T}\sum_{\mathcal{T}} a_k] \to L$, and the common pairwise covariance satisfies $\mathrm{Var}(\frac{1}{T}\sum_{\mathcal{T}} a_k) = \frac{1}{T}\mathrm{Var}(a_1) + \frac{T-1}{T}\mathrm{Cov}(a_1, a_2) \to 0$, which forces $\mathrm{Cov}(a_1, a_2) \to 0$ (the first term is $O(1/T)$). Then $\mathrm{Var}(\frac{1}{T_0}\sum_{\mathcal{T}_0} a_k) = \frac{1}{T_0}\mathrm{Var}(a_1) + \frac{T_0-1}{T_0}\mathrm{Cov}(a_1, a_2) \to 0$, so the subset average also concentrates on L. □

A.6 Value of a design (Theorem 1)

Proof. Fix a nested sequence $D(K) = (N, T, T_0)$ satisfying Assumption 1. Decompose the target as $y_0 = \boldsymbol{x}_{0,\mathcal{T}_0}^\top \boldsymbol{\beta}_{\mathcal{T}_0} + \sum_{k\notin\mathcal{T}_0} \beta_k x_{0k} + \eta_0$. Conditioning on $\boldsymbol{d} = (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{T}_0})$, the observed covariates $\boldsymbol{x}_{0,\mathcal{T}_0}$ are known, the unobserved x_{0k} ($k \notin \mathcal{T}_0$) are centered and independent of everything else, and η_0 is independent and centered. The three blocks are conditionally uncorrelated, so

$$\mathrm{Var}(y_0 \mid \boldsymbol{d}) = \underbrace{\boldsymbol{x}_{0,\mathcal{T}_0}^\top \Sigma_{\mathcal{T}_0} \boldsymbol{x}_{0,\mathcal{T}_0}}_{\text{(a)}} + \underbrace{\frac{\sigma_x^2}{K}\sum_{k\notin\mathcal{T}_0} \mathbb{E}[\beta_k^2 \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}]}_{\text{(b)}} + \underbrace{\sigma_\eta^2\sigma_\beta^2}_{\text{(c)}}, \tag{8}$$

where $\Sigma_{\mathcal{T}_0} \equiv \mathrm{Var}(\boldsymbol{\beta}_{\mathcal{T}_0} \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ and I used $\mathrm{Var}(x_{0k}) = \sigma_x^2/K$. We take the three limits in turn; recall $\mathrm{Var}(y_0) = \sigma_\beta^2(\sigma_x^2 + \sigma_\eta^2)$.

(b). By Lemma 5 with $\mathcal{A}_K = \{1, \dots, K\} \setminus \mathcal{T}_0$ (of size $K - T_0$), $\frac{1}{K}\sum_{k\notin\mathcal{T}_0} \mathbb{E}[\beta_k^2 \mid \cdot] \xrightarrow{p} (1 - t_0)\sigma_\beta^2$, so (b)$\xrightarrow{p} \sigma_x^2(1 - t_0)\sigma_\beta^2$.

(a). Set $a_k \equiv \mathrm{Var}(\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ for $k \in \mathcal{T}$. These are exchangeable (the law of $(\boldsymbol{\beta}_{\mathcal{T}}, \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ is invariant under permuting trained indices) and L^2-bounded ($\mathbb{E}a_k^2 \le \mathbb{E}[\mathbb{E}[\beta_k^4 \mid \cdot]] = 3\sigma_\beta^4$). By Lemma 4 and Proposition 7, $\frac{1}{T}\sum_{k\in\mathcal{T}} a_k = \frac{1}{T}\mathrm{tr}\,\Sigma_{\mathcal{T}} \xrightarrow{p} \sigma_\beta^2(1 - \mathcal{I}(n, t))$; Lemma 6 then gives $\frac{1}{T_0}\mathrm{tr}\,\Sigma_{\mathcal{T}_0} = \frac{1}{T_0}\sum_{k\in\mathcal{T}_0} a_k \xrightarrow{p} \sigma_\beta^2(1 - \mathcal{I}(n, t))$. Next, $\boldsymbol{x}_{0,\mathcal{T}_0} \sim \mathcal{N}(0, \frac{\sigma_x^2}{K} I_{T_0})$ is independent of the training sample, so conditionally on it the quadratic form (a) has mean $\frac{\sigma_x^2}{K}\mathrm{tr}\,\Sigma_{\mathcal{T}_0}$ and variance $2(\sigma_x^2/K)^2\,\mathrm{tr}(\Sigma_{\mathcal{T}_0}^2) \le 2(\sigma_x^2/K)^2\sigma_\beta^4 T_0 = O(1/K)$. Hence (a)$= \frac{\sigma_x^2}{K}\mathrm{tr}\,\Sigma_{\mathcal{T}_0} + o_p(1) = \sigma_x^2\frac{T_0}{K}\cdot\frac{1}{T_0}\mathrm{tr}\,\Sigma_{\mathcal{T}_0} + o_p(1) \xrightarrow{p} \sigma_x^2 t_0\,\sigma_\beta^2(1 - \mathcal{I}(n, t))$.

Combine. Subtracting (8) from $\mathrm{Var}(y_0)$,

$$\hat{v}_{D(K)}(\boldsymbol{d}) \xrightarrow{p} \sigma_\beta^2(\sigma_x^2 + \sigma_\eta^2) - \sigma_x^2 t_0\sigma_\beta^2(1 - \mathcal{I}) - \sigma_x^2(1 - t_0)\sigma_\beta^2 - \sigma_\eta^2\sigma_\beta^2 = \sigma_\beta^2\sigma_x^2\,t_0\,\mathcal{I}(n, t) = \mathcal{V}(n, t, t_0).$$

Finally, the preceding decomposition also gives convergence of expectations. The terms in (8) are uniformly integrable: (a) is controlled by $0 \preceq \Sigma_{\mathcal{T}_0} \preceq \sigma_\beta^2 I$, (b) by Lemma 5, and (c) is

constant. Therefore

$$v(D(K)) = \mathrm{Var}(y_0) - \mathbb{E}[\mathrm{Var}(y_0 \mid \boldsymbol{d})] \to \mathcal{V}(n, t, t_0).$$

□

A.7 Signal-share representation (Proposition 2)

Proof. Signal. $\mathcal{S}_K = \frac{1}{T}\sum_{k\le T}\sum_{i\le N} x_{ik}^2$ has mean $\frac{N}{K}\sigma_x^2 \to n\sigma_x^2$ and variance $2\sigma_x^4\,\frac{N}{TK^2} \to 0$, so $\mathcal{S}_K \xrightarrow{p} \mathcal{S}(n) = n\sigma_x^2$.

Noise. With $u_0 = y_0 - \boldsymbol{x}_{0,\mathcal{T}}^\top \boldsymbol{\mu}$ and $\delta \equiv \boldsymbol{\beta}_{\mathcal{T}} - \boldsymbol{\mu}$, write $u_0 = \varepsilon_0 + \boldsymbol{x}_{0,\mathcal{T}}^\top \delta$ where $\varepsilon_0 = \sum_{k\notin\mathcal{T}} \beta_k x_{0k} + \eta_0$. Conditional on training, these two pieces are uncorrelated; the first has normalized variance $\to \sigma_\varepsilon^2(t)$ (its conditional variance is governed by the same omitted-scale as s^2, cf. Lemma 3), and the second, since $\boldsymbol{x}_{0,\mathcal{T}} \sim \mathcal{N}(0, \frac{\sigma_x^2}{K} I_T)$ is independent of δ, has normalized variance $\frac{\sigma_x^2}{K\sigma_\beta^2}\operatorname{tr}\Sigma_{\mathcal{T}} = t\sigma_x^2 \cdot \frac{1}{T\sigma_\beta^2}\operatorname{tr}\Sigma_{\mathcal{T}} \xrightarrow{p} t\sigma_x^2(1-\mathcal{I})$ by Lemma 4. Hence, $\mathcal{E}_K \xrightarrow{p} \mathcal{E}(\mathcal{I}; t) = \sigma_\varepsilon^2(t) + t\sigma_x^2(1-\mathcal{I})$.

Fixed point. With $\Psi(\mathcal{I}; n, t) = \mathcal{S}(n)/(\mathcal{S}(n) + \mathcal{E}(\mathcal{I}; t))$ and $\sigma_x^2 = 1$, the equation $\mathcal{I} = \Psi(\mathcal{I})$ is $\mathcal{I}\,(n + 1 + \xi - t\mathcal{I}) = n$, i.e. $t\mathcal{I}^2 - (n + 1 + \xi)\mathcal{I} + n = 0$, which is (7). Thus $\mathcal{I}(n, t)$ is the unique fixed point in $[0, 1]$. □

A.8 Comparative statics (Corollary 1, Propositions 3–4)

Proof. Normalize $\sigma_x^2 = 1$, set $m \equiv 1+\xi$ and $S \equiv \sqrt{(n+m)^2 - 4nt}$, so that by (7) $S = n+m-2t\mathcal{I}$. Implicit differentiation of (7) yields

$$\mathcal{I}_n = \frac{1-\mathcal{I}}{S} > 0, \qquad \mathcal{I}_t = \frac{\mathcal{I}^2}{S} > 0. \tag{9}$$

Since $V = t_0\mathcal{I}$, this gives $V_n, V_t > 0$ and, from $S_n = 1 - 2t\mathcal{I}_n$ and $S_t = -2\mathcal{I} - 2t\mathcal{I}_t$,

$$\mathcal{I}_{nn} = -\frac{2(1-\mathcal{I})\left(n + m - t(1+\mathcal{I})\right)}{S^3} < 0, \qquad \mathcal{I}_{tt} = \frac{2\mathcal{I}^3\,(2S + t\mathcal{I})}{S^3} > 0,$$

(the factor $n + m - t(1+\mathcal{I}) = S + t(1-\mathcal{I}) > 0$), establishing strict concavity in n (Corollary 1) and strict convexity in t (Proposition 3).

For Proposition 4, $V_{nt_0} = \mathcal{I}_n > 0$ and $V_{tt_0} = \mathcal{I}_t > 0$ by (9), so target scope complements both training margins. The sign of $V_{nt} = t_0\mathcal{I}_{nt}$ is the content. Differentiating $\mathcal{I}_n = (1-\mathcal{I})/S$ in t,

$$\mathcal{I}_{nt} = \frac{\mathcal{I}(2-3\mathcal{I})\,S + 2t\mathcal{I}^2(1-\mathcal{I})}{S^3},$$

whose sign is that of $h(\mathcal{I}; t, m) \equiv \mathcal{I}(2-3\mathcal{I})S + 2t\mathcal{I}^2(1-\mathcal{I})$. Using $S = (m - t\mathcal{I}(2-\mathcal{I}))/(1-\mathcal{I})$ and $n = \mathcal{I}(m - t\mathcal{I})/(1-\mathcal{I})$ from (7), and dividing by the positive $\mathcal{I}/(1-\mathcal{I})$, the sign equals

that of

$$H(\mathcal{I}; t, m) \equiv m(2 - 3\mathcal{I}) + t\mathcal{I}\,(-2 + 4\mathcal{I} - \mathcal{I}^2).$$

Now $H(\frac{2}{3}; t, m) = \frac{4t}{27} > 0$, $H(1; t, m) = t - m \leq 0$, and on $\mathcal{I} \in (\frac{2}{3}, 1)$, $H_{\mathcal{I}} = -3m + t(-2 + 8\mathcal{I} - 3\mathcal{I}^2) < 0$ (since $-2 + 8\mathcal{I} - 3\mathcal{I}^2 < 3 \leq 3m$). Hence H has a unique root $q^\star(t, \xi) \in (\frac{2}{3}, 1)$, and $H > 0$ below it. As $\mathcal{I} = \mathcal{I}(n, t)$ strictly increases in n, define $\widehat{n}(t; \xi) \equiv q^\star(m - tq^\star)/(1 - q^\star)$ (the value of n at which $\mathcal{I} = q^\star$); then

$$V_{nt} \gtreqless 0 \iff \mathcal{I}_{nt} \gtreqless 0 \iff \mathcal{I} \lesseqgtr q^\star \iff n \lesseqgtr \widehat{n}(t; \xi).$$

At the threshold, the parametrization $q = q^\star$ gives $m = \dfrac{tq(4q - q^2 - 2)}{3q - 2}$ and $\widehat{n} = \dfrac{tq^3}{3q - 2}$, $q \in (\frac{2}{3}, 1)$, from which $\widehat{n}$ is increasing in m (hence in ξ) and decreasing in t. □

A.9 Proof of Proposition 5

Proof. For part (i), fix t and define

$$h(n) \equiv \mathcal{I}(n, t).$$

By Corollary 1, h is strictly concave, and $h(0) = \mathcal{I}(0, t) = 0$. Let $a, b > 0$. Since

$$a = \frac{a}{a + b}(a + b) + \frac{b}{a + b} \cdot 0,$$

strict concavity gives

$$h(a) > \frac{a}{a + b} h(a + b).$$

The same argument gives

$$h(b) > \frac{b}{a + b} h(a + b).$$

Adding the two inequalities yields

$$h(a) + h(b) > h(a + b).$$

Setting $a = n_1$ and $b = n_2$, and multiplying by $t_0 > 0$, proves part (i).

For part (ii), fix n and define

$$g(s) \equiv s\,\mathcal{I}(n, s).$$

Then $g(0) = 0$. By Proposition 3, $\mathcal{I}(n, \cdot)$ is strictly increasing and strictly convex. Hence

$$g''(s) = 2\,\mathcal{I}_t(n, s) + s\,\mathcal{I}_{tt}(n, s) > 0,$$

so g is strictly convex. Let $a, b > 0$. Since

$$a = \frac{a}{a + b}(a + b) + \frac{b}{a + b} \cdot 0,$$

strict convexity gives

$$g(a) < \frac{a}{a+b}\, g(a+b),$$

and similarly

$$g(b) < \frac{b}{a+b}\, g(a+b).$$

Adding the two inequalities yields

$$g(a) + g(b) < g(a+b).$$

Setting $a = t_1$ and $b = t_2$ proves part (ii). □

B Robustness: Simulation Evidence

The closed form in Theorem 1 relies on three properties of the DGP introduced in Section 2.1.1: the covariates are independent, homoskedastic, and Gaussian. The proof uses these properties only insofar as they imply that the empirical spectral distribution of $\boldsymbol{X}_{N,\mathcal{T}}^{\top}\boldsymbol{X}_{N,\mathcal{T}}/N$ converges to the Marchenko–Pastur law, which delivers the closed-form expression for $\mathcal{I}(n,t)$. In this section, I use Monte Carlo simulation to (i) verify that the theorem holds under the baseline DGP at moderate K, and (ii) document how value departs from the closed form when each assumption is relaxed in turn.

Although the main analysis restricts attention to nested designs, this simulation keeps the target covariate set fixed and varies training scope. Covariates that are observed at prediction time but not in training cannot be used, so the effective target scope is the overlap between training and target covariates. Under the canonical ordering used in the simulation, this overlap is $\min\{t, t_0\}$. The simulated value is therefore compared to

$$V(n, t, \min\{t, t_0\}) = \min\{t, t_0\}\mathcal{I}(n, t).$$

The region $t < t_0$ should be read as a dominated, non-nested design: the agent observes some target covariates whose weights have not been trained. It is included only to show how value changes as training scope catches up with a fixed target scope.

Validation under the baseline DGP. Figure 2 plots the realized value $\hat{V}(\boldsymbol{d})$, averaged over 100 Monte Carlo replications at $K = 800$, against the relevant closed-form prediction. In the left panel this is $\mathcal{V}(n, t, t_0)$. In the right panel, where target scope is held fixed while training scope varies, it is $\mathcal{V}(n, t, \min\{t, t_0\})$. The left panel sweeps the training scale n at fixed $t = t_0 = 0.5$; the right panel sweeps the training scope t at fixed $n = 2$, $t_0 = 0.5$. Simulated values track theory across both margins, including the kink at $t = t_0$, where training scope catches up with the fixed target scope.

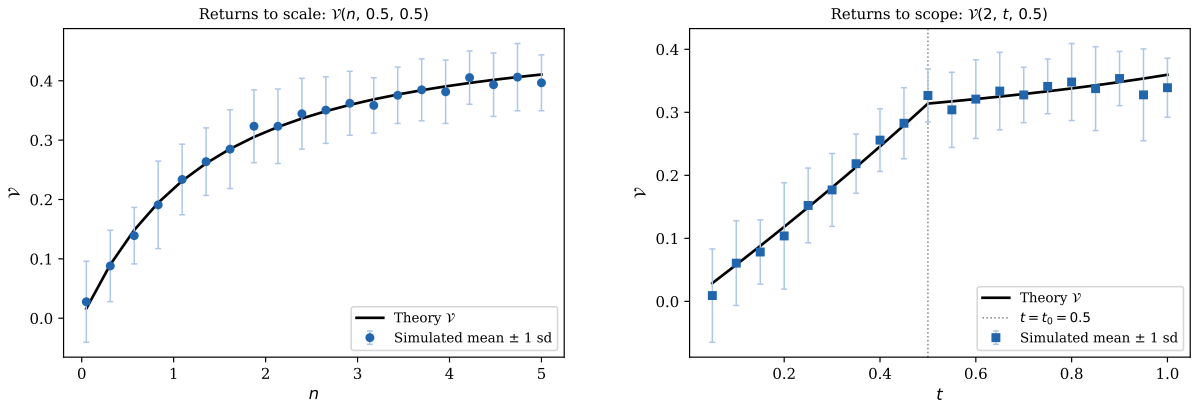


Figure 2: Simulated and theoretical value. Left: value as training scale varies. Right: value as training scope varies, holding target scope fixed at $t_0 = 0.5$. For $t < t_0$, only the overlap between training and target covariates is useful, so effective target scope is $\min\{t, t_0\}$. The kink at $t = t_0$ marks the point at which all target covariates are also included in training.

Convergence in K. Figure 3 documents the speed of convergence to the asymptotic limit as K grows, holding $(n, t, t_0) = (2, 0.6, 0.4)$ and $\xi = 0.5$ fixed. Under the baseline DGP, the distribution of $\hat{v}(\boldsymbol{d})$ concentrates around $\mathcal{V}(n, t, t_0)$ already at $K = 200$, and the interquartile range contracts steadily thereafter. The heavy-tailed variant (t_3 idiosyncratic shocks) displays the same convergence pattern with a modest increase in dispersion at small K, suggesting that finite second moments are enough for concentration in this design. Correlated covariates ($\rho = 0.6$, Toeplitz) also converge to the i.i.d. limit, consistent with the bulk-spectrum result documented in Figure 4. By contrast, the heteroskedastic variant ($\alpha = 1$, linear variance decay) converges to a limit approximately three times larger than the i.i.d. prediction: the realized value stabilizes near 0.75 as $K \to \infty$ while the dashed line sits at $\mathcal{V} \approx 0.25$. This is not sampling noise — the gap is a genuine shift in the asymptotic limit induced by the linear-decay schedule, consistent with the analysis in Figure 4.

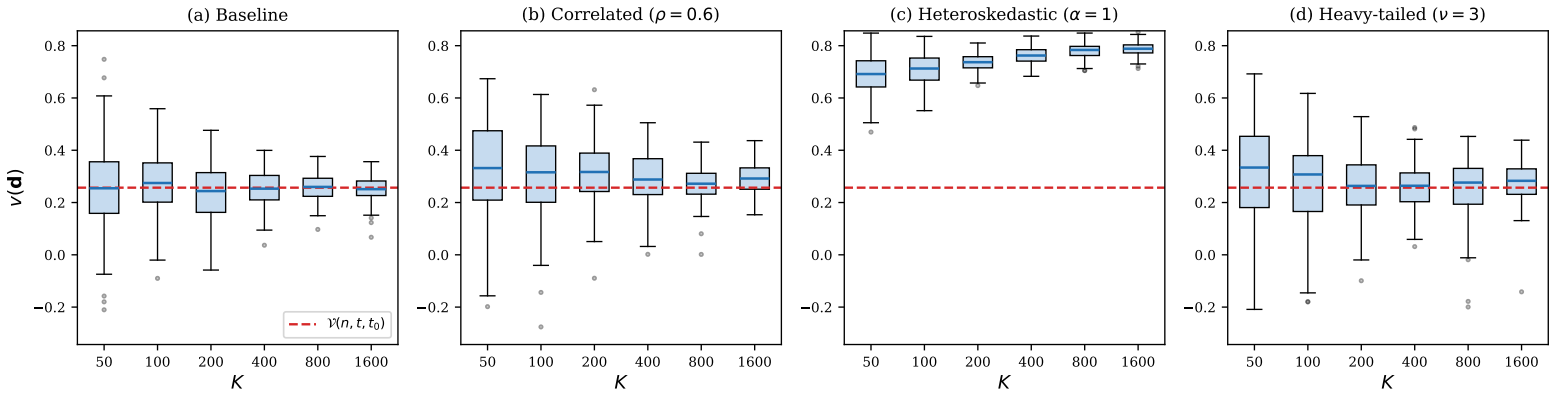


Figure 3: Distribution of realized dataset value $\hat{v}(\boldsymbol{d})$ across $R = 100$ Monte Carlo replications as K grows, at $(n, t, t_0) = (2.0,\ 0.6,\ 0.4)$ and $\xi = 0.5$. Each box spans the interquartile range; the dashed red line is the closed-form limit $\mathcal{V}(n, t, t_0)$ from Theorem 1. Panels (a)–(d) correspond to the baseline, Toeplitz-correlated ($\rho = 0.6$), heteroskedastic ($\alpha = 1$), and heavy-tailed (t_3) DGPs respectively.

Departures from the closed form. I next relax each assumption in turn, holding $t = t_0 = 0.5$ and $\xi = 0.5$ and sweeping n. Figure 4 reports the resulting realized values.

Correlation. Local correlation in the form of an AR(1) covariance, $\Sigma_{jk} \propto \rho^{|j-k|}$, leaves the limit essentially unchanged for $\rho \in \{0.3, 0.7\}$: simulated values lie on the i.i.d. theory curve. The empirical spectral distribution of a Toeplitz Σ remains in the bulk regime, and the Marchenko–Pastur limit deforms continuously to a known image. Equicorrelation ($\Sigma_{jk} = \rho$ for $j \neq k$) is qualitatively different: it introduces an eigenvalue of order K that violates the bulk regime, and the realized value sits systematically above the i.i.d. limit.

Heteroskedasticity. The departure is even larger and changes sign across schedules. A linear decay of per-covariate variances raises realized value far above the i.i.d. limit, while an 80/20 spike—in which a small share of covariates carry most of the variance—lowers it. Both schedules preserve the trace condition $\sum_k s_k^2/K \to \sigma_x^2$, so the gap is not a normalization artifact. Intuitively, the agent's effective dimension shrinks when signal concentrates on a small subset of indices: when those indices are within the training scope, learning is easier; when they straddle the boundary $\mathcal{T}$, the trained covariates carry only a fraction of the total signal and value falls.

Heavy tails. Replacing the Gaussian idiosyncratic shock with a Student-t_ν shock at $\nu = 3$ leaves the limit unchanged (Figure 3, panel (d)): the simulation suggests that the same limit is stable in this design when shocks have finite variance, although the formal proof in Appendix A uses Gaussian conjugacy.

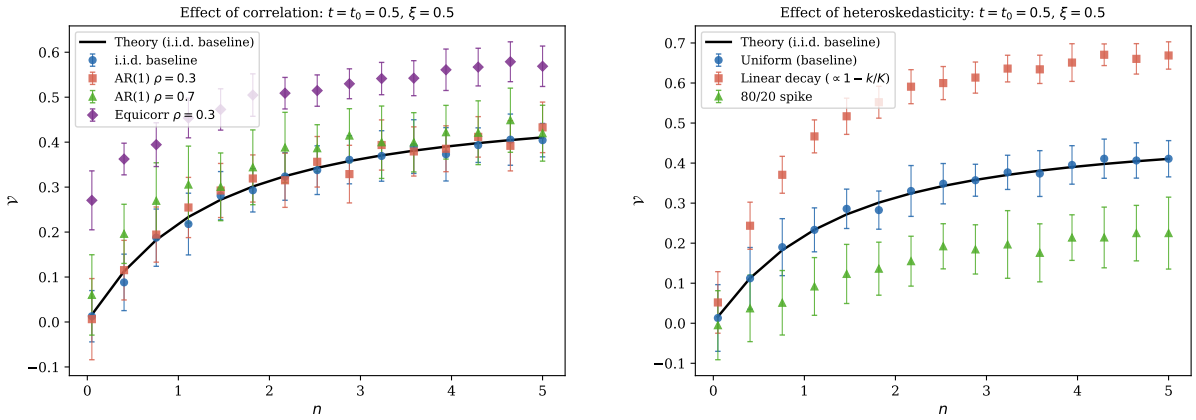


Figure 4: Realized $\hat{v}(\boldsymbol{d})$ versus n at $K = 800$, $t = t_0 = 0.5$, $\xi = 0.5$, under departures from the baseline DGP. Left: correlated covariates. Right: heteroskedastic covariates with a preserved trace condition. The black curve is the i.i.d. closed form $\mathcal{V}(n, 0.5, 0.5)$.

Interpretation. The simulation evidence delineates the scope of Theorem 1. The closed form is robust to local correlation in covariates and to heavy-tailed shocks, but not to features that concentrate the spectrum of Σ—either through a dominant eigenvalue (equicorrelation) or through a heterogeneous diagonal (heteroskedasticity with mass on a small subset of indices). In economic terms, the closed form prices a dataset whose covariates are roughly equally informative; when predictive power is concentrated on a small subset, the value of the dataset depends on whether the agent's training scope captures that subset, which is a separate design margin not parameterized by (n, t, t_0) alone. Extending $\mathcal{V}$ to non-bulk spectra is left for future work.